MIL-S-12472 (ORD)
13 February 1953
SUPERSEDING
FXS-1066 (in part)
24 February 1951

MILITARY SPECIFICATION

SYNCHROS, 60 CYCLE, 31-AND 23-SIZE

1. SCOPE

1.1 Scope.- The word synchro is a term, analogous to a trade name, used by the Military Establishment. It applies to certain components of self synchronous systems. This specification covers 115-volt, 60-cycle snychros of the types listed in Table I.

1.2 CLASSIFICATION

1.2.1 Code for Type Designations of 115-Volt Synchros.- 115-Volt synchros are classified according to size, function, and power supply frequency. This information is contained in the type designation. The term "115-volt synchro" includes the following:

a. Synchros which are excited from a 115 volt supply.
b. Synchros excited from other synchros which are energized from a 115 volt supply.

1.2.2 Size.- The first two digits designate the maximum diameter in tenths of an inch. If the diameter is not exactly a whole number of tenths, the next higher tenth is used.

1.2.3 Function.- The succeeding group of letters indicate the function in accordance with the following table:

First Letter	Function
C	Control
T	Torque

Succeeding Letters	Function
D	Differential
R	Receiver
T	Transformer
X	Transmitter

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:51Z
Check the source to verify that this is the current version before use.

1.2.4 The succeeding digit indicates the frequency of the power supply in accordance with the following table:

Supply Frequency (cps)	Code
60	6

1.2.5 Modifications.---A lower case letter such as "a", "b", etc., following the frequency digit indicates a modification of the basic design that affects the external mechanical dimensions or the electrical characteristics.

1.2.6 Experimental Modifications.---A code enclosed in parenthesis ( ), following the type designation described above, indicates an experimental unit as follows:

First Letter	Meaning
X	Experimental
Second Letter	
A	Air Force
N	Navy
W	Army
Succeeding Digits	
1, 2, 3, etc.	1st, 2nd, 3rd, etc. modifications

1.2.7 Illustration.---23TDX6 indicates a 115 volt 60 cycle differential Torque transmitter with an overall diameter of 2.250 inches.

1.3 Definitions of Types.---

1.3.1 Transmitter, Synchro - (torque) (TX).---A unit, the rotor of which is mechanically positioned, for transmitting electrical information corresponding to angular positions of the rotor. It is constructed for operation with torque receivers (TR) and torque differential receivers (TDR). In certain cases it may also be used as a control transmitter (CX).

1.3.2 Transmitter, Synchro - (control) (CX).---A unit, the rotor of which is mechanically positioned, for transmitting electrical information corresponding to angular positions of the rotor. The electrical outputs are similar to those of the torque transmitters(TX), but are used only to supply control transformers (CT) or control differential transmitters (CDX).

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:51Z
Check the source to verify that this is the current version before use.

1.3.3 Transmitter, Synchro - (torque differential) (TDX).--- A unit, the rotor of which is mechanically positioned, for modifying electrical angular information received from a torque transmitter (TX) and transmitting electrical information corresponding to the sum or difference of the electrical input angle and its rotor position angle. This unit is designed for operation with torque receivers (TR).

1.3.4 Transmitter, Synchro - (control differential) (CDX).--- A unit, the rotor of which is mechanically positioned, for modifying electrical angular information received from a control transmitter (CX) and transmitting electrical information corresponding to the sum or difference of the electrical input angle and its rotor position angle. This unit is designed for operation with control transformers (CT).

1.3.5 Receiver, Synchro - (torque) (TR).--- A unit, the rotor of which is free to turn, which when energized will assume a position in accordance with the electrical information received. This unit is designed for operation with torque transmitters (TX) and torque differential transmitters (TDX).

1.3.6 Receiver, Synchro (torque differential) (TDR).--- A unit the rotor of which is free to turn, which will assume a position in accordance with the sum or difference of the electrical angular information received from two synchro transmitters (TX).

1.3.7 Transformer, Synchro Control (CT).--- A unit the primary of which is supplied with electrical angular information, and the secondary of which supplies a potential proportional to the sine of the difference between the electrical input angle and the control transformer rotor angle. This unit is designed for operation with control transmitters (CX) and control differential transmitters (CDX).

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS, AND PUBLICATIONS.

2.1 The following specifications, standard, and drawings, of the issue in effect on date of invitation for bids, form a part of this specification.

SPECIFICATIONS

Military

MIL-S-901	-	Shockproof-Equipment, Class HI(high impact); and tests for (Shipboard Application).
MIL-E-5272	-	Environmental Testing, Aeronautical and Associated Equipment, (General Specification for).
MIL-P-12134	-	Packaging of Synchros, A.C., Types 15, 23, and 31-60 and 400 cycles.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:51Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS (Continued)

U. S. Army

51-7 - Instruments, Fire Control; General Specification Governing the Manufacture and Inspection of.

Ordnance Department Tentative

FXS-973 - Vacuum Impregnation Procedure for Electrical Windings and Coil Lamination Assemblies.

STANDARD
MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS

The drawings listed in Table I shall form a part of this specification.

(Copies of specifications, standard and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer).

TABLE I

115 Volt 60 Cycle Synchros

Type	Type Designation	Ordnance Outline Drawing (1)
Transmitter, Synchro-(torque)	31TX6	C76673h3
Receiver, Synchro-(torque)	31TR6	C76673h2
Transmitter, Synchro- (torque differential)	31TDX6	
Receiver, Synchro- (torque differential)	31TDR6S1 *	C76673h5
Transmitter, Synchro-(torque)	23TX6a	C8212778
	23TX6	C7676354
Receiver, Synchro-(torque)	23TR6a	C8212781
	23TR6	C7676355
Transmitter, Synchro- (torque differential)	23TDX6a	C8212797
	23TDX6	C7676360
Transmitter, Synchro-(Control)	23CX6a	C8212794
	23CX6	C7676359

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:51Z
Check the source to verify that this is the current version before use.

Type	Type Designation	Ordnance Outline Drawing
Transformer, Synchro-(Control)	23CT6a 23CT6	C8212789 C7676357
Transmitter, Synchro-(Control differential)	23CDX6a 23CDX6	C8212786 C7676356

NOTE:

* The suffix S1 in Type 31TDR6S1 does not conform to the code specified in para. 1.2. The S1 is used to designate pigtail leads.

3. REQUIREMENTS

3.1 Synchros shall be manufactured in accordance with the requirements of this specification, and with the drawings referred to in 2.1. The synchros shall be able to meet the following operating requirements.

3.1.1 Operating Speed.---Except as indicated in para. 4.22.1.1, all synchros covered by this specification shall be capable of operating at 1200 rpm continuously for 1000 hours, when being driven by straight spur gears or in any other way which does not apply an axial load, and still meet the requirements of this specification. Brush wear may become noticeable if the synchros are operated in excess of 1000 hours at 1200 rpm.

3.1.2 Ambient Temperature.---All synchros covered by this specification shall sustain no damage while operating or standing in an ambient temperature of -80° F to +160°F.

3.1.3 Vibration and Shock.---The synchros covered by this specification shall be able to operate satisfactorily after being subjected to any combination of vibration and shock stresses described in 4.20 and 4.21.

3.1.4 Corrosion Resistance and Humidity.---The materials called for in the drawings of 2.1 are corrosion resistant except for the laminations in certain machines and the bearings. The laminations depend on the coating for corrosion protection and the bearings on the rust inhibitor of the lubricant. The synchros are intended to be used preferably in an enclosed clean dry space which is protected against the entry of moisture, dirt, dust, or insects. However, if the synchro is carefully manufactured, the rotor and stator assemblies (with the bearings omitted) should be able to undergo without rusting or corroding the cycling humidity test which is specified in MIL-E-5272 Procedure I and outlined in 4.24.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:51Z
Check the source to verify that this is the current version before use.

3.1.5 Insulation Resistance.---The synchros shall have an insulation resistance of at least 10 megohms between each winding and the case and between the primary and secondary windings after being subjected to a high potential stress of 900 volts (nominal) RMS 60 cycles.

3.1.6 System Errors.---In addition to meeting the accuracy requirements of 3.7 and 3.8, the stator windings of all synchros shall be phased so that there is a positive peak of the second harmonic error in the 0° - 120° sector. Thus the synchro errors of the transmitter and control transformers in any synchro system will in most cases subtract provided that the stator connections are S1 to S-1, S-2 to S-2 and S-3 to S-3. This means that in any system the direction of rotation of transmitter and the control transformers as viewed from the shaft extension end should be the same for minimum system error.

3.2 All synchros having the same type designation shall be mechanically and electrically interchangeable in standard applications.

3.3 Synchros shall be identified by nameplates as called for by the applicable drawings of 2.1. The serial numbers on the nameplates shall be assigned by the custodian of the applicable drawings. Blocks of numbers will be assigned for each type to the contractors in such a manner that no two synchros of a given type shall have the same serial number. The serial numbers shall occur on the synchros approximately in the chronological order of manufacture. It shall be permissible to have numbers unused in the sequence when the nameplates are damaged or when synchros fail to pass inspection.

3.4 Definitions and Conventions.---

3.4.1 Direction of Rotation.---Clockwise or counterclockwise rotation is determined when facing the shaft extension end of the synchro.

3.4.2 Angular Position.---The angular position of the synchro is the angular displacement of its rotor from the electrical zero position measured counterclockwise.

3.4.3 Primary and Secondary Windings.---The primary winding is the one which receives energizing power from either the supply line (as in transmitters and torque receivers) or from another synchro (as in control transformers). In the case of differential transmitters and differential receivers, the primary winding is the outer winding, which has less turns than the inner winding.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:51Z
Check the source to verify that this is the current version before use.

3.4.4 Electrical Angle.---The electrical angle is the angle "a" which satisfies the relative magnitudes and phase of the secondary voltage of a synchro transmitter in accordance with the following vector equations:

$$\left.\begin{array}{l} E(S13) = NE(R21)\sin a \\ E(S32) = NE(R21)\sin(a \pm 120^\circ) \\ E(S21) = NE(R21)\sin(a \pm 240^\circ) \end{array}\right.$$

where N is the ratio between the maximum rms voltage between the secondary terminals and the primary voltage.

E(R21) is the voltage between terminals R2 and R1.
E(S13) is the voltage between terminals S1 and S3.
Other voltages are similarly defined.

3.4.5 Units.---Unless otherwise specified the units for
Angles are degrees, minutes, and seconds
Potential is volts rms.---------Impedance is ohms.
Current is amperes rms.

3.4.6 Phase.---The time phase at a point in a synchro system is the phase of the voltage at that point with respect to the energizing voltage of the synchro system. The time phase is measured in degrees.

3.4.7 Electrical Zero.---

3.4.7.1 Transmitters and Receivers.---The electrical zero position of transmitters and receivers is that position of the rotor with respect to the stator in which minimum voltage is induced in the secondary circuit S1-S3 and in which the secondary voltages E(S12) and E(S32) are approximately in phase with the primary voltage E(R21). The minimum voltage is defined as the zero point of the in-phase voltage of the fundamental frequency. Electrical Zero Test of Transmitter and Receivers shall be as outlined in 4.4.2.

3.4.7.2 Differential Transmitters and Differential Receivers.---The electrical zero position of differential transmitters and differential receivers is that position of the rotor with respect to the stator in which minimum voltage is induced in circuit R1-R3 and in which the voltages E(R32) and E(R21) are approximately in-phase with the corresponding voltages E(S32) and E(S21) when the synchro is energized by applying 0.866 times the rated voltage between terminal S2 and terminal S1 which is connected to S3. The minimum voltage is defined as the zero point of the in-phase voltage of the fundamental frequency. Electrical Zero Tests of Differential Transmitters and Differential Receivers shall be as outlined in 4.4.3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:51Z
Check the source to verify that this is the current version before use.

3.4.7.3 Control Transformers---The electrical zero position of a control transformer is that position of the rotor with respect to the stator in which minimum voltage is induced in the secondary circuit R1-R2 when the unit is energized by applying 0.866 times the rated voltage between terminal S2 and terminal S1 which is connected to S3. It is so determined that for small deflections of the rotor counterclockwise from electrical zero the induced voltage E(R12) is approximately in-phase with the voltage E(S21) or E(S23). The minimum voltage is defined as the zero of the in-phase voltage of the fundamental frequency. Electrical Zero Test of Control Transformers shall be as outlined in 4.4.4.

3.5 Workmanship.---Workmanship shall be in accordance with the highest grade instrument practice.

3.6 Information on Drawings.---

3.6.1 The drawings of 2.1 contain complete information of the material, the dimensions and the finish of each part of the synchro.

3.6.2 The drawings of 2.1 do not provide complete information on the method of inserting the insulation and windings, the cementing technique, the many precautions that must be taken in assembly, or controls required in processing magnetic materials. These items and in general all manufacturing processes, techniques and skills, which form part of the "know-how" of instrument makers, may not be covered by the drawings. The manufacturer may choose his production process so long as the product is in accordance with the drawings and this specification.

3.7 Limiting Values for 115 Volt Synchros.--(For tests which apply to 31TDR6S1 see Table II and III.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:51Z
Check the source to verify that this is the current version before use.

3.7.1 115 Volt Torque Transmitters (TX).---

Tolerances: Energizing Current ± 10%
Energizing Power ± 20%
Trans. Ratio ± 1%

	Units	Sizes 23	Sizes 31	Ref. Para.
Primary voltage (nominal)	volts	115	115	
Energizing primary current	milliamps	210	420	
Energizing power	watts	5	6.5	
Transformation ratio		0.783	0.783	4.7
No load temperature rise (max)	°C	60	60	4.15
Electrical error (max) 1/	minutes	8	8	4.5(Q)4/ 4.26(I)4/
Phase Rotation		Included in Electrical Error Measurement		4.5 4.26
Null voltage, total (max) 2/	millivolts	125	125	4.9
Fund. component of null voltage (max) 2/ 3/	millivolts	75	75	4.9
Torque gradient	oz-in/deg.	0.13±.01	.33±.03	4.12
Friction torque Room Temp. (maximum) -55°C	oz-in	0.2	0.2	4.13
	oz-in	0.5	0.5	4.13
Radiation (max)	microvolts	2.5 microvolts from .15 - 65mc 5.0 microvolts from 65 - 150mc		4.14

Note: No unit shall spin when energized under extreme conditions of power supply at any displacement from synchronism, when fitted with the proper dial shown on figure 20.

1/ The first positive peak of the second harmonic of the error curve shall be in the 0 - 120 degree sector provided there is a second harmonic error of more than 3 minutes maximum value.

2/ As read on a vacuum tube voltmeter indicating the average value of the voltage wave in terms of the rms value of an equivalent sine wave. This value is true fundamental voltage, having been corrected for the attenuation of the filter used.

3/ This value is the true fundamental voltage, having been corrected for the attenuation of the filter used.

4/ See 4.1 for meaning of Q and I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:51Z
Check the source to verify that this is the current version before use.

3.7.2 115 Volt Torque Receivers (TR).---

Tolerances: Energizing Current ± 10%
Energizing Power ± 20%
Trans. Ratio ± 1%

	Units	Sizes 23	Sizes 31	Ref. Para.
Primary voltage (nominal)	volts	115	115	
Energizing primary current	milliamps	210	420	
Energizing power	watts	5	6.5	
Transformation ratio		0.783	0.783	4.7
No load temp. rise (max)	°C	60	60	4.15
Receiver error (max)	minutes	60	48	4.6(Q)2 4.22(I)
Electrical error (max) 1/	minutes	8	8	4.5(Q) 4.26(I)
Phase Rotation		Included in Electrical Error Measurement		4.5(Q) 4.26(I)
Torque gradient	oz-in/deg.	0.13±.01	.40±.03	4.12
Synchronizing time 30°	sec	1	1	4.17
(maximum) 179°	sec	2	2	4.17
Radiation (max)	microvolts	2.5 microvolts from .15 - 65mc 5.0 microvolts from 65 - 150mc		4.14

Note: No unit shall spin when energized under extreme conditions of power supply at any displacement from synchronism, when fitted with the proper dial shown on figure 20.

1/ The first positive peak of the second harmonic of the error curve shall be in the 0 - 120 degree sector provided there is a second harmonic error of more than 3 minutes maximum value.

2/ See 4.1 for meaning of Q and I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:51Z
Check the source to verify that this is the current version before use.

3.7.3 115 Volt Torque Differential Transmitters (TDX).--- Tolerances: Energizing Current ± 10%
Energizing Power ± 20%
Trans. Ratio ± 1%

	Units	Sizes 23	Sizes 31	Ref. Para.
Max. primary voltage (nominal)	volts	90	90	
Energizing current	milliamps	380	550	
Energizing power	watts	11	11	
Transformation ratio		1.154	1.154	4.7
No load temp. rise (max)	°C	60	45	4.15
Electrical error rotor	minutes	8	8	4.5(Q)2/ 4.26(I)
(maximum) stator 1/	minutes	8	8	4.5.5(Q)2/ 4.26(I)
Phase Rotation		Included in Electrical Error Measurement		4.5.(Q) 4.26(I)
Torque gradient	oz-in/deg.	.034±.004	.15±.01	4.12
Friction Torque room temp.	oz-in	0.2	0.2	4.13
(maximum) -55°C	oz-in	0.5	0.5	4.13
Radiation (max)	microvolts	2.5 microvolts from 15 - 65mc 5.0 microvolts from 65 - 150mc		4.14

Note: No unit shall spin when energized under extreme conditions of power supply at any displacement from synchronism, when fitted with the proper dial shown on figure 20.

1/ The first positive peak of the second harmonic of the error curve shall be in the 0 - 120 degree sector provided there is a second harmonic error of more than 3 minutes maximum value (stator only).

2/ See 4.1 for meaning of Q and I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:51Z
Check the source to verify that this is the current version before use.

3.7.4 115 Volt Torque Differential Receivers (TDR).--- Tolerances: Energizing Current ± 10%
Energizing Power ± 20%
Trans. Ratio ± 1%

	Units	Sizes 23	Sizes 31	Ref. Para.
Max. primary voltage (nominal)	volts	90	90	
Energizing current	milliamps	380	550	
Energizing power	watts	11	11	
Transformation ratio		1.154	1.154	4.7
No load temp. rise (max)	°C	60	45	4.15
Receiver error (max)	minutes	60	48	4.6(Q) 4.28(I)
Electrical error rotor 4/	minutes	8	8	4.5(Q)2/
(maximum) stator 1/ 4/	minutes	8	8	4.26(I)
Phase Rotation 4/		Included in Electrical Error Measurement		4.5(Q) 2/ 4.26(I)
Sensitivity 3/—				4.11
Torque gradient	oz-in/deg.	.034±.004	.15±.01	4.12
Synchronizing time 30° 4/	sec	1	1	4.17
(maximum) 179° 4/	sec	2	2	4.17
Radiation (max)	microvolts	2.5 microvolts from .15 - 65mc 5.0 microvolts from 65 - 150mc		4.14

Note: No unit shall spin when energized under extreme conditions of power supply at any displacement from synchronism, when fitted with the proper dial shown on figure 20.

1/ The first positive peak of the second harmonic of the error curve shall be in the 0 - 120 degree sector provided there is a second harmonic error of more than 3 minutes maximum value (stator only).

2/ See 4.1 for meaning of Q and I.

3/ Test to be performed only on Type 31TDR6S1

4/ Tests do not apply to Type 31TDR6S1

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:51Z
Check the source to verify that this is the current version before use.

3.7.5 115 Volt Control Transmitters (CX).---

Tolerances: Energizing Current ± 10%
Energizing Power ± 20%
Trans. Ratio ± 1%

	Units	Size 23	Ref. Para.
Primary voltage (nominal)	volts	115	
Energizing primary current	milliamps	75	
Energizing power	watts	1.6	
Transformation ratio		0.783	4.7
Null voltage, total (max) 1/	millivolts	100	4.9
Fund. component of null voltage (max) 2/ 1/	millivolts	75	4.9
No load temp. rise (max)	°C	25	4.15
Electrical error (max) 3/	minutes	8	4.5(Q)4/ 4.26(I)
Phase Rotation		Included in Electrical Error Measurement	4.5(Q)4/ 4.26(I)
Electrical Stability	minutes	0.25	4.10
Friction torque room temp.	oz-in	0.2	4.13
(maximum) -55°C	oz-in	0.5	4.13
Radiation (max)	microvolts	2.5 microvolts from .15 - 65mc 5.0 microvolts from 65 - 150mc	4.14

1/ As read on a vacuum tube voltmeter indicating the average value of the voltage wave in terms of the rms value of an equivalent sine wave.

2/ This value is the true fundamental voltage, having been corrected for the attenuation of the filter used.

3/ The first positive peak of the second harmonic of the error curve shall be in the 0 - 120 degree sector provided there is a second harmonic error of more than 3 minutes maximum value.

4/ See 4.1 for meaning of Q and I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:51Z
Check the source to verify that this is the current version before use.

3.7.6 115 Volt Control Differential Transmitters (CDX).--- Tolerances: Energizing Current ± 10%
Energizing Power ± 20%
Trans. Ratio ± 1%

	Units	Size 23	Ref. Para.
Max. primary voltage (nominal)	volts	90	
Energizing Current	milliamps	102	
Energizing power	watts	2.2	
Transformation ratio		1.154	4.7
Null voltage, total (max) 1/	millivolts	100	4.9
Fund. component of null voltage (max) 2/ 1/	millivolts	75	4.9
No load temp. rise (max)	°C	25	4.15
Electrical error rotor	minutes	8	4.5(Q) 4/ 4.26(I)
(maximum) stator 3/	minutes	8	4.5(Q)
Phase Rotation		Included in Electrical Error Measurement	4.26(I)
Electrical Stability	minutes	0.25	4.10
Friction Torque room temp.	oz-in	0.2	4.13
(maximum) -55°C	oz-in	0.5	4.13
Radiation (max)	microvolts	2.5 microvolts from .15 - 65mc 5.0 microvolts from 65 - 150mc	4.14

1/ As read on a vacuum tube voltmeter indicating the average value of the voltage wave in terms of the rms value of an equivalent sine wave.

2/ This value is the true fundamental voltage, having been corrected for the attenuation of the filter used.

3/ The first positive peak of the second harmonic of the error curve shall be in the 0 - 120 degree sector provided there is a second harmonic error of more than 3 minutes maximum value.

4/ See 4.1 for meaning of (Q) and I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:51Z
Check the source to verify that this is the current version before use.

3.7.7 115 Volt Control Transformers (CT).---

Tolerances: Energizing Current ± 10%
Energizing Power ± 20%
Trans. Ratio ± 1%

	Units	Size 23	Ref. Para.
Max primary voltage (nominal)	volts	90	
Energizing Current	milliamps	20	
Energizing power	watts	0.50	
Transformation ratio		0.735	4.7
Null voltage, total (max) 1/	millivolts	60	4.9
Fund. component of null voltage (max) 2/	millivolts	40	4.9
Electrical error (max) 3/	minutes	6	4.5(Q) 4/ 4.26(I) 4/
Phase Rotation		Included in Electrical Error Measurement	4.5(Q) 4.26(I)
Electrical Stability	minutes	0.25	4.10
Friction torque room temp.	oz-in	0.2	4.13
(maximum) -55°C	oz-in	0.5	4.13
Radiation (max)	microvolts	2.5 microvolts from .15 -65mc 5.0 microvolts from 65 - 150mc	4.14

1/ As read on a vacuum tube voltmeter indicating the average value of the voltage wave in terms of the rms value of an equivalent sine wave.

2/ This value is the true fundamental voltage, having been corrected for the attenuation of the filter used.

3/ The first positive peak of the second harmonic of the error curve shall be in the 0 - 120 degree sector provided there is a second harmonic error of more than 3 minutes maximum value.

4/ See 4.1 for meaning of Q and I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:51Z
Check the source to verify that this is the current version before use.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Tests to determine whether the synchros have been made in accordnance with the drawings of 2.1 and this specification are divided into two categories:

Qualification Tests (designated by Q)
Inspection Tests (designated by I)

4.2 Qualification (Q) Tests.---Q tests shall be performed at a government laboratory on a sample of four synchros. The government reserves the right to subject samples of synchros to the Q tests at any time during the life of the production contract, and to reject the production if these samples fail to pass. In this case the sample shall be selected in accordance with instructions of the procuring agency. Unless otherwise specified all tests shall be performed at 22°C ± 5°C and at atmospheric pressure.

4.2.1 For acceptance 115 volt synchros shall pass the tests listed in Table II.

TABLE II

Qualification Tests of 115 Volt Synchros

Qualification Test	Functional Classification CX	CDX	CT	TX	TR	TDX	TDR	31 TDR6S1	Ref. Para.
Electrical Error	Q	Q	Q	Q	Q	Q	Q		4.5
Receiver Error					Q		Q	Q	4.6
Transformation Ratio	Q	Q	Q	Q	Q	Q	Q	Q	4.7
Open Circuit Current and Power	Q	Q	Q	Q	Q	Q	Q	Q	4.8
Null Voltage	Q	Q	Q	Q					4.9
Phase Rotation	Q	Q	Q	Q	Q	Q	Q		4.5
Electrical Stability	Q	Q	Q						4.10
Sensitivity								Q	4.11
Torque Gradient				Q	Q	Q	Q	Q	4.12
Maximum Friction Torque	Q	Q	Q	Q		Q			4.13
Radiation	Q	Q	Q	Q	Q	Q	Q		4.14
Temperature Rise	Q	Q		Q	Q	Q	Q	Q	4.15
Spinning					Q		Q		4.16
Synchronizing Time					Q		Q		4.17
High Potential and Insulation Resistance	Q	Q	Q	Q	Q	Q	Q	Q	4.18
Dielectric	Q	Q	Q	Q	Q	Q	Q	Q	4.19
Vibration	Q	Q	Q	Q	Q	Q	Q	Q	4.20
Shock	Q	Q	Q	Q	Q	Q	Q	Q	4.21
Endurance	Q	Q	Q	Q	Q	Q	Q	Q	4.22
Humidity	Q	Q	Q	Q	Q	Q	Q	Q	4.24
Mechanical Inspection	Q	Q	Q	Q	Q	Q	Q	Q	4.25

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:51Z
Check the source to verify that this is the current version before use.

4.3 Inspection (I) Tests.---I tests shallbe performed at the place of manufacture under the supervision of the government inspector. Unless otherwise specified all tests shall be performed at 22°C ±5°C and at atmospheric pressure. The procuring agency will furnish acceptance inspection instructions. Attribute inspection in general shall be conducted.

4.3.1 Production Lots.---Production lots of synchros or component parts shall consist of units of product which have been made in a single plant over a period of continuous production during which no significant changes in process have occurred, and in which homogeniety within the lot may reasonably be expected. Production lot sizes shall be as specified by the procuring agency.

4.3.2 Inspection Lots.---The inspection lot, which is a collection of units of product submitted by a supplier for Government Inspection, does not necessarily coincide in size with a production lot. The inspection lot shall be defined, formed and presented as specified in Standard MIL-STD-105.

4.3.3 For acceptance 115 volt synchros shall pass the tests listed in Table III.

TABLE III

Inspection Tests of 115 Volt Synchros

	Functional Classification								
Inspection Test	CX	CDX	CT	TX	TR	TDX	TDR	31 TDR6S1	Ref. Para.
Electrical Error	I	I	I	I	I	I	I		4.26
Receiver Error					I		I	I	4.28
Transformation Ratio	I	I	I	I	I	I	I	I	4.29
Open Circuit Current and Power	I	I	I	I	I	I	I	I	4.30
Null Voltage	I	I	I	I					4.31
Phase Rotation	I	I	I	I	I	I	I		4.26
Sensitivity								I	4.27
Spinning					I		I		4.32
Synchronizing Time					I		I		4.33
Maximum Friction Torque	I	I	I	I		I			4.34
High Potential and Insulation Resistance	I	I	I	I	I	I	I	I	4.35
Dielectric	I	I	I	I	I	I	I	I	4.36
Mechanical Inspection	I	I	I	I	I	I	I	I	4.37

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:51Z
Check the source to verify that this is the current version before use.

4.3.4 In addition to the tests listed in Table III, the synchros shall be subject to any other tests specified herein which the contracting officer deems necessary to determine conformance with this specification.

4.4 Electrical Zero.---Measurements of all angular displacements of the rotors of synchros shall be referred to a standard position which will be designated as "Electrical Zero". The rotor shall first be examined to insure that it is well balanced and free to rotate without excessive friction. Caution shall be taken in testing units without dampers to prevent their spinning. External friction may be applied or the applied voltage made very low at first and then increased to normal after the rotor has come to rest.

4.4.1 Coarse Electrical Zero Marking.---Electrical zero shall be approximately located to within 10° by matching the arrow stamped on the frame with the index mark on the shaft extension.

4.4.2 Electrical zero of 115 volt transmitters and torque receivers.---The Electrical zero position of a 115 volt receiver or transmitter shall be determined by connecting the primary and secondary as shown in figure 1.

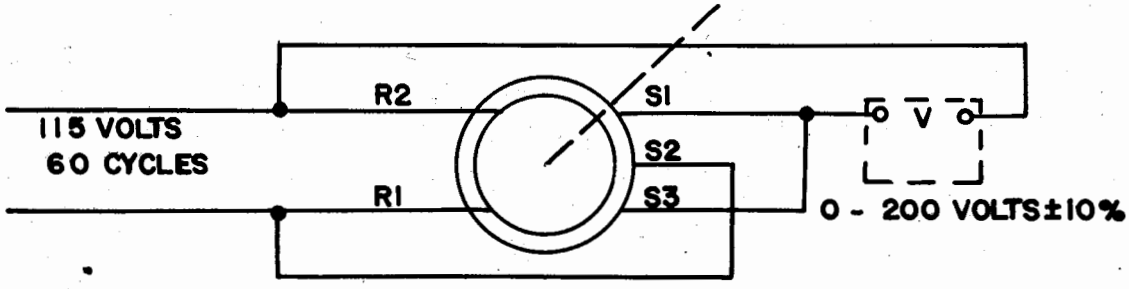


FIGURE 1

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:51Z
Check the source to verify that this is the current version before use.

The synchro shaft will rotate to either of two positions which are 180° apart. The correct approximate position is that one which produces the smaller voltage reading in V. The electrical zero position should then be accurately determined as follows. Without rotating the synchro shaft, reconnect synchro as shown in figure 2.

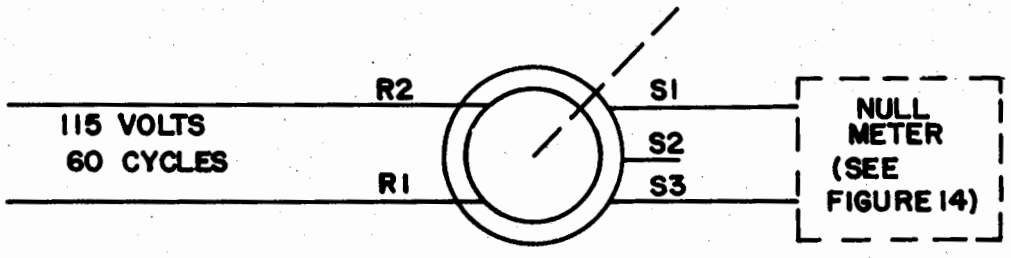


FIGURE 2

Rotate the Synchro shaft through the smaller angle that will produce a zero reading on the nullmeter. That position is the electrical zero position.

4.4.3 Electrical zero of 115 volt differential transmitters and differential receivers.---The electrical zero position of a 115 volt differential receiver or transmitter shall be determined by connecting the synchro as shown in figure 3.

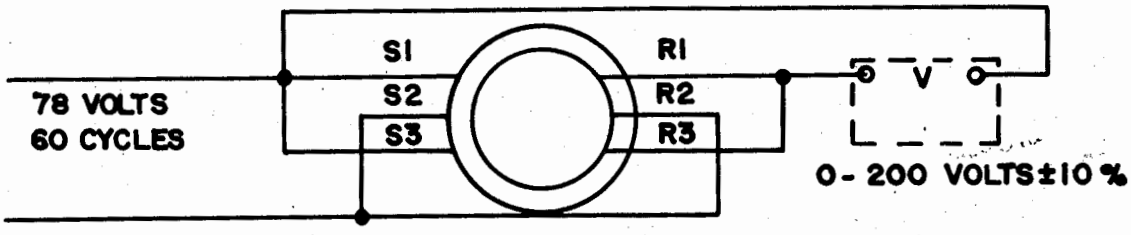


FIGURE 3

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:51Z
Check the source to verify that this is the current version before use.

The synchro shafts of units with slip rings will rotate to either of two positions which are 180° apart. The correct approximate position is the one which produces the smallest voltage reading in V. The synchro shafts of units with pigtails instead of slip rings will lock in place approximately midway between the limit stops. The correct approximate position is the one which produces a voltage of less than 50 volts in V. The electrical zero position should then be accurately determined as follows. Without rotating the shaft, reconnect the synchro as shown in figure 4.

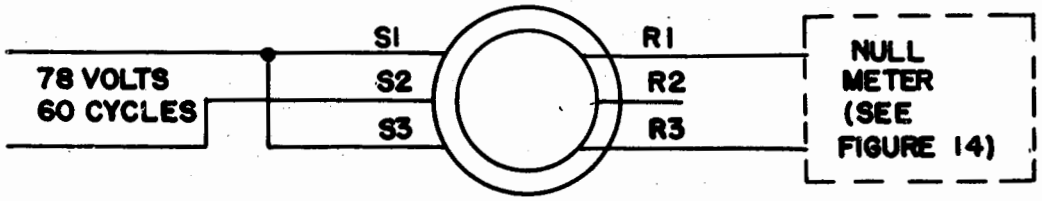


FIGURE 4

Rotate the synchro shaft through the smallest angle that will produce a zero reading of the nullmeter. That position is the electrical zero position.

4.4.4 Electrical zero of 115 Volt Control Transformers.---The electrical zero position of a 115 volt control transformer shall be determined by connecting the synchro as shown in figure 5.

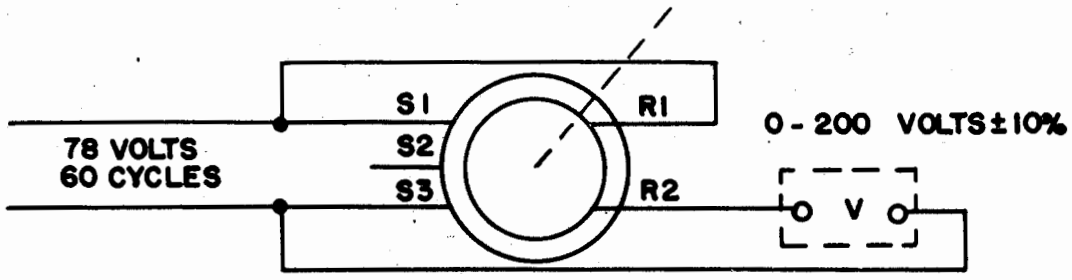


FIGURE 5

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:51Z
Check the source to verify that this is the current version before use.

The electrical zero position is approximately at the minimum voltage position. This position should be accurately located by connecting the synchro as shown in figure 6.

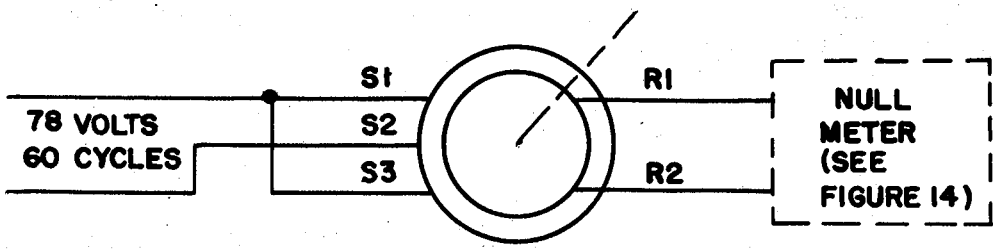


FIGURE 6

Rotate the synchro shaft through the smallest angle that will produce a zero reading on the nullmeter. That position is the electrical zero position.

4.5 Electrical Error Qualification Test.---

4.5.1 Except for 31TDR6S1, all synchros shall be tested for electrical error by the proportional voltage method. Synchro receivers and differential receivers in addition shall be tested for receiver error against transmitters. Control synchros shall be tested for electrical error at -55 ± 5°C, +22 ±5°C and +75 ± 5°C. Torque synchros shall be tested at -55 ± 5°C, +22 ± 5°C and +55 ± 5° C.

4.5.2 The errors shall be determined for at least every ten (10) degree position of the rotor. Torque receivers shall be subject to test with the shaft in any of the following positions: 0°, 45°, 90°, 135°, 180°, from the vertical.

4.5.3 Before testing the synchro for electrical error by the proportional voltage method, the electrical zero position of the synchro under test shall be accurately determined as described in 4.4.2, 4.4.3, or 4.4.4, as applicable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:51Z
Check the source to verify that this is the current version before use.

4.5.4 Proportional Voltage Method.---Electrical Error (Q) Test for Transmitters and Receivers--This method consists in comparing the actual position of the synchro rotor with the electrical position of the synchro. The electrical position of the synchro is obtained from the following equations:

$$\text{Electrical position} = \phi \pm (N \pm 3M)60^{\circ}$$

$$R = \tfrac{1}{2} - \tfrac{1}{2}\sqrt{3}\cot(\phi \pm 60^{\circ})$$

$$R = \frac{\text{one of the smaller secondary voltages}}{\text{largest secondary voltage}}$$

M = 0 when voltage E(S13) is in phase with voltage E(R21).
M = 1 when voltage E(S13) is 180° out of phase with voltage E(R21).

and N is determined from the following tables:

Terminals giving largest secondary voltage	Terminals giving small secondary voltage	Value of N
S2-S1	S1-S3	0
S1-S3	S3-S2	1
S3-S2	S2-S1	2

The basic circuit employed for electrical error testing of synchro transmitters is shown in Figure 7. The unit shall be energized with rated voltage 60 cycles across R1-R2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:51Z
Check the source to verify that this is the current version before use.

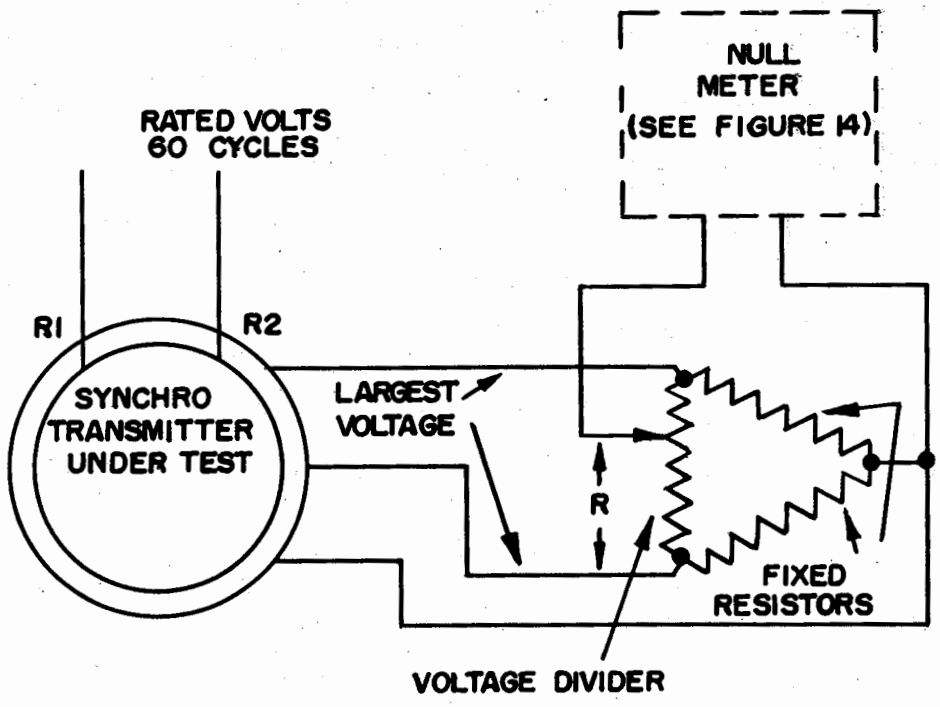


FIGURE 7

Basic Circuit Employed for Electrical Error (Q) Test of Synchro Transmitters and Receivers.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:51Z
Check the source to verify that this is the current version before use.

Two 10,000 ohm resistors and a non-inductive decade voltage divider of 10,000 ohms shall be connected in delta across the output circuits S1, S2 and S3. The resistors and the decade voltage divider shall be equal to within 0.01%. The decade voltage divider shall be linear to within 0.01% and produce a phase shift of no more than 10 minutes. The voltage divider shall be connected across the secondary terminals giving the largest voltage, and the resistors shall be connected across the terminals giving the smaller voltages. The variable tap on the voltage divider shall be connected to the common terminal of the two fixed resistors through a null meter (fig. 14). The voltage divider shall be set at the proper ratio (R in the formula) for the desired electrical position and the synchro shaft shall be turned until a zero reading of the nullmeter is obtained. The position of the rotor is then recorded. The error at a given rotor position is defined as the rotor position minus the electrical position. To facilitate the settings of the voltage divider a table of values of ϕ corresponding to various values of R is given in Table IV.

Table IV

Values of R for Various Values of Angle

ϕ	R
0	.00000
5	.09617
10	.18479
15	.26795
20	.34730
25	.42423
30	.50000
35	.57577
40	.65270
45	.73205
50	.81521
55	.90383
60	1.00000

4.5.4.1 The maximum errors, in absolute value, of 115 volt control transmitters shall not exceed those given in 3.7.5. The maximum errors, in absolute value, of torque transmitters shall not exceed those given in 3.7.1. The maximum errors of torque receivers shall not exceed those given in 3.7.2. In addition, a positive maximum of the second harmonic of the error shall occur in the 0° to 120° sector provided there is a second harmonic error of more than 3 minutes maximum value.

4.5.4.2 Phase Rotation.--The phase rotation is automatically correct if the values for electrical error measurement in Table IV agree.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:51Z
Check the source to verify that this is the current version before use.

4.5.5 Electrical Error (Q) Test for Control Transformers.---The basic circuit for electrical error testing of control transformers is shown in Figure 8. The stator terminals of the control transformer shall be connected to the corresponding stator terminals of a size 23 synchro control transmitter operated at rated voltage and frequency. The output terminals R1-R2 of the control transformer shall be connected to the nullmeter. The energizing transmitter shall be turned until its electrical position, measured as described in 4.5.4 is at the desired value. The angular position of the unit under test that produces a zero reading of the nullmeter shall be determined. The error at a given rotor position is the rotor position minus the electrical position. The maximum errors (in absolute value) of control transformers shall not exceed the limits given in 3.7.7. In addition, a positive maximum value of the second harmonic shall occur in the 0° to 120° sector provided there is a second harmonic error of more than 3 minutes maximum value.

4.5.6 Electrical Error(Q) Test for Differential Transmitters and Differential Receivers.---With the exception of Type 31TDR6S1, the electrical error of both the rotor and the stator shall be determined as described in 4.5.6.1 and 4.5.6.2 below.

4.5.6.1 Electrical Error (Q) Test for Differential Transmitter and Torque Differential Receiver Rotors.---The basic circuit for electrical error testing of the rotor windings of differential synchros is shown in Figure 9. The unit shall be energized with 78 volts cycles across terminal S2 and terminal S1 which is connected to S3. The electrical position of the rotor is obtained from the following equation:

$$\text{Electrical position} = \emptyset \pm (N \pm 3M)60^{\circ}$$

where $\emptyset$ is determined as described in 4.5.4.

M = 0 when voltage E(R13) is in phase with voltage E(S23).
M = 1 when voltage E(R13) is 180° out of phase with voltage E(S23).
and N is obtained from the following table:

Terminals giving largest secondary voltage	Terminals giving small secondary voltage	Value of N
R3-R2	R1-R3	0
R1-R3	R2-R1	1
R2-R1	R3-R2	2

The error at a given rotor position is equal to the rotor position minus the electrical position. The maximum errors (in absolute value) of control differential transmitters shall not exceed the limits given in 3.7.6. The maximum errors (in absolute value) of torque differential transmitters shall not exceed the limits given in 3.7.3. The maximum errors (in absolute value) of torque differential receivers shall not exceed the limits given in 3.7.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:51Z
Check the source to verify that this is the current version before use.

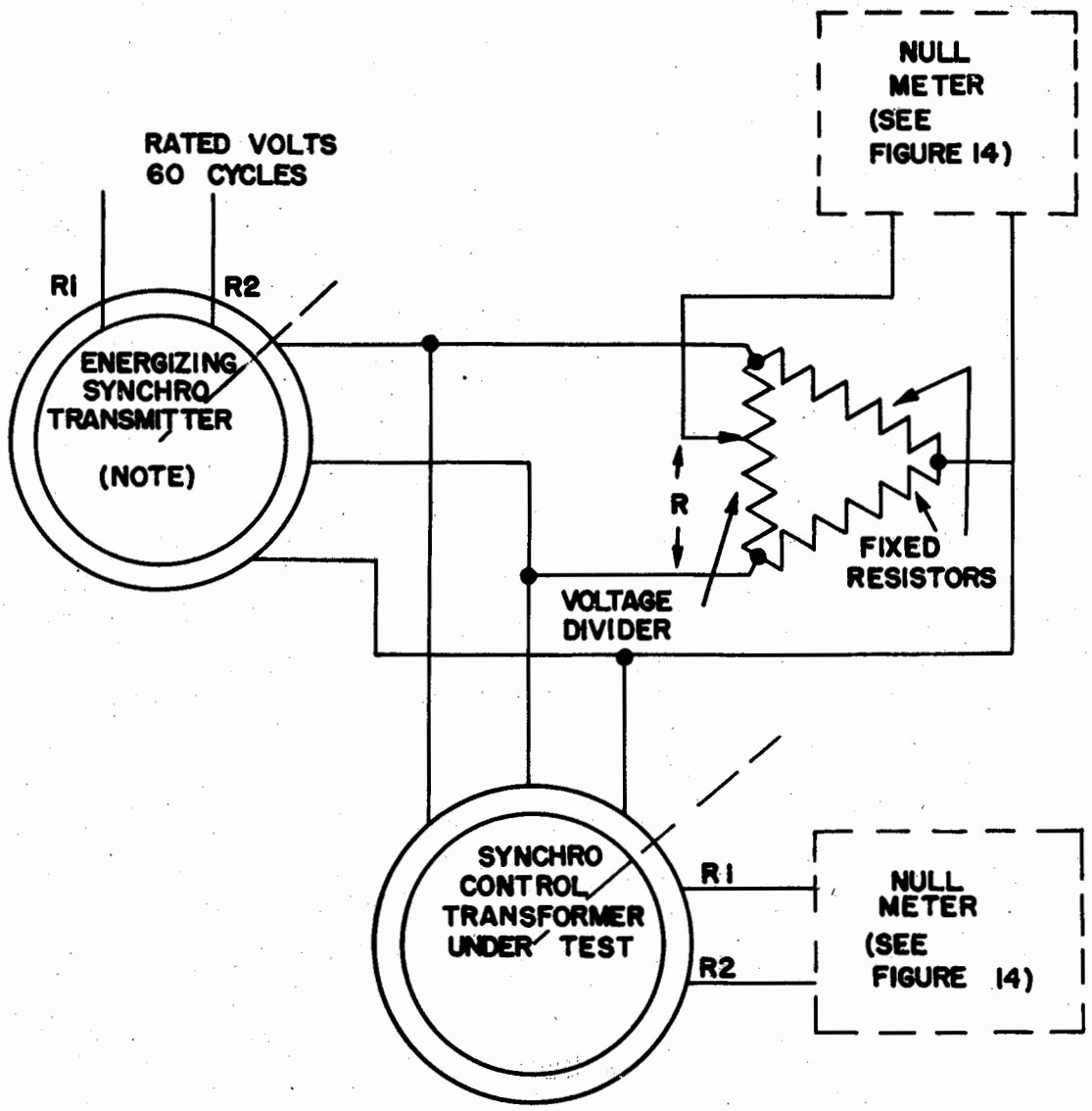


FIGURE 8

Basic Circuit Employed for Electrical Error (Q) Test of Control Transformers.

Note: This transmitter shall have passed the electrical error test of 4.5.5. The measured results of this test do not include in any way the errors of this transmitter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:51Z
Check the source to verify that this is the current version before use.

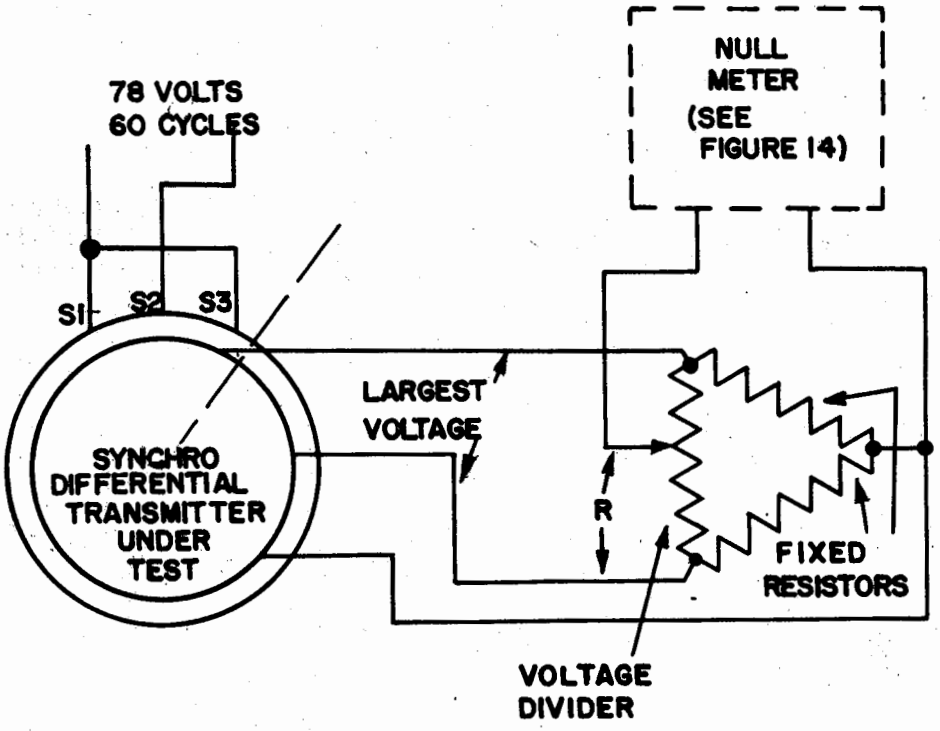


FIGURE 9

Basic Circuit Employed for Electrical Error (Q) Test of the Rotor windings of Differential Transmitters.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:51Z
Check the source to verify that this is the current version before use.

4.5.6.2 Electrical Error(Q) Test for Differential Transmitter and Torque Differential Receiver Stators.---The basic circuit for electrical error testing of the stator windings of differential transmitters is shown in Figure 10. The stator terminals of the synchro under test shall be connected to the corresponding stator terminals of a synchro transmitter of equal or larger size, operated at rated voltage and frequency. Rotor terminals R1-R3 of the differential transmitter shall be connected to a nullmeter. The energizing transmitter shall be turned until its electrical position, measured as described in 4.5.4 is at the desired value. The angular position of the unit under test that produces a zero reading of the nullmeter shall be determined. The error at a given rotor position is defined as the rotor position minus the electrical position. The maximum errors (in absolute value) of control differential transmitters shall not exceed the limits given in 3.7.6. The maximum errors (in absolute value) of torque differential transmitters shall not exceed the limits given 3.7.3. The maximum errors (in absolute value) of torque differential receivers shall not exceed the limits given in 3.7.4. In addition, a positive maximum of the 2nd harmonic shall occur in the 0° to 120° sector provided there is a second harmonic error of more than 3 minutes maximum value.

4.6 Receiver Error Qualification Test.---

4.6.1 Receiver Error (Q) Test for Torque Receivers.---Torque Receivers shall be tested against torque transmitters of equal size. The receiver error shall be determined for both clockwise and counterclockwise rotations. The receiver errors, corrected for the calibration errors, shall not exceed those given in 3.7.2.

4.6.2 Receiver Error (Q) Test for Torque Differential Receivers.--- Except for type 31TDR6S1 all other sizes of torque differential receivers shall be tested against two torque transmitters of the same size one connected to the rotor of the differential receiver and the other connected to the stator of the differential receiver. Two complete sets of error measurements in each direction of rotation shall be taken.

4.6.2.1 One transmitter shall be locked on electrical zero and the setting of the other transmitter shall be varied in not more than ten (10) degree steps recording the angular position assumed by the differential receiver.

4.6.2.2 The above test shall be repeated but with the other transmitter clamped on electrical zero and the setting of the transmitter varied in not more than (10) degree steps as before.

4.6.2.3 The errors, corrected for the calibration errors, shall not exceed those given in 3.7.4.

4.6.3 Receiver Error (Q) Test for 31TDR6S1.---The unit shall be tested against two size 31 torque transmitters that have satisfactorily met the

Source: https://assist.dla.mil - Downloaded: 2026-10-17T22:51Z
Check the source to verify that this is the current version before use.

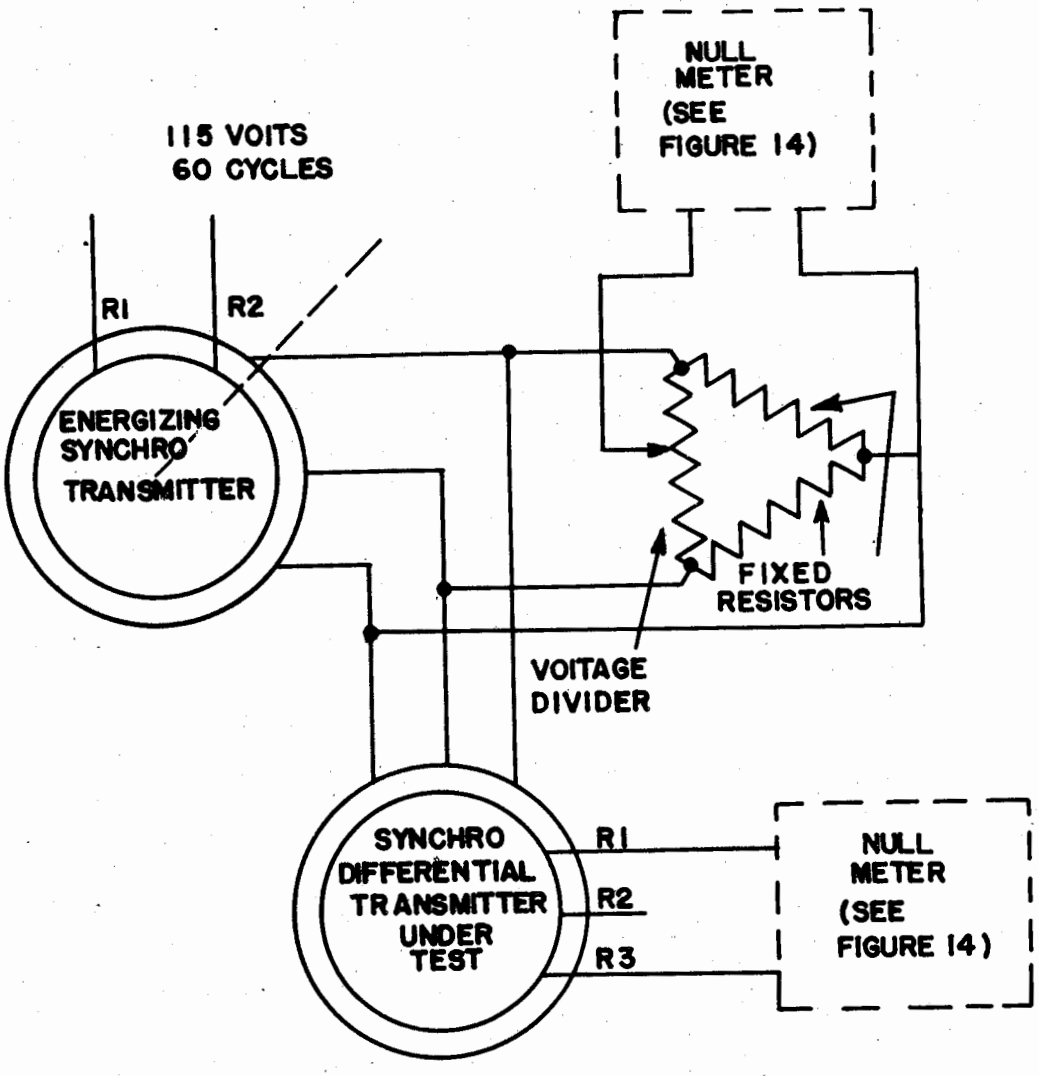


FIGURE 10

Basic Circuit Employed for Electrical Error (Q) Test of the Stator Windings of Differential Transmitters.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:51Z
Check the source to verify that this is the current version before use.

requirements of this specification. The torque transmitters shall be mechanically connected back to back such that, with rotors locked, they can be rotated simultaneously. The stator leads of the torque transmitters shall be connected to the unit under test in the following manner: S1 to S1, S2 to S2, S3 to S3 and S1 to R1, S2 to R2, S3 to R3. The torque transmitters shall be energized by applying 115 volts 60 cycles to the rotors.

4.6.3.1 Both transmitter rotors shall be locked on electrical zero and transmitters shall be rotated at 1 rpm for at least four revolutions in each direction. The maximum displacement of the rotor shaft, of the torque differential receiver under test, from electrical zero is the receiver error. The error shall not exceed that given in 3.7.4.

4.7 Transformation Ratio Qualification Test.- By the transformation ratio of a synchro is meant the ratio of the no-load maximum secondary voltage to the supply voltage applied to the primary. For this test the voltage measuring device shall not alter the open circuit secondary voltage by more than 0.1 percent.

4.7.1 115-Volt Transmitters and Receivers.- The transformation ratio of 115 volt transmitters and receivers (which should be 90/115=0.783) shall be obtained by applying 115 volts ± 1 percent at 60 cycles ± 1 percent across synchro terminals R1-R2 and turning the rotor until voltage E (S13) is a maximum. The transformation ratio shall be 0.783 ± 1 percent.

4.7.2 115-Volt Differential Transmitters and Torque Differential Receivers.- Except for 31TDR6S1, the transformation ratio of 115 volt differential transmitters and differential receivers (which should be 90/78 = 1.154) shall be obtained by applying 78 volts ± 1 percent at 60 cycles ± 1 percent across terminal S2 and terminal S1 which is connected to S3. The rotor shall be turned until voltage E(R13) is a maximum. The transformation ratio shall be 1.154 ± 1 percent.

4.7.2.1 Transformation Ratio for 31TDR6S1. - The leads of Type 31TDR6S1 shall be disconnected and tested in accordance with 4.7.3.

4.7.3 115-Volt Control Transformers.- The transformation ratio of 115 volt control transformers (which should be 57.3/78 = 0.735) shall be obtained by applying 78 volts ± 1 percent at 60 cycles ± 1 percent across terminal S2 and terminal S1 which is connected to S3. The rotor shall be turned until voltage E (R12) is a maximum. The transformation ration shall be 0.735 ± 1 percent.

4.8 Open Circuit Current and Power (Q) of 115 Volt Synchros.- The input current and power to the 115 volt synchro shall be measured with the synchro energized at rated voltage and frequency and with the secondary leads open. For differential synchros and control transformers the applied voltage is 78 volts, at 60 cycles, which shall be applied across terminals S2 and S1 which is connected to S3. The measurements

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:51Z
Check the source to verify that this is the current version before use.

shall be made with the synchro at normal operating temperature. The primary current and power shall not exceed the values given in 3.7. The open circuit current or power shall not vary by more than 2 percent as the rotor is turned through a complete revolution.

4.9 Null Voltage (Q).---The null voltage and the fundamental component of the null voltage shall be determined by employing the circuit of Figure 11. These voltages shall be read on a vacuum tube voltmeter indicating the average value of the voltage wave in terms of the rms value of an equivalent sine wave. The synchro shall be connected as indicated in the table of Figure 11. The null voltage readings shall be made with the synchro at the electrical angles shown in the table. With the switch in position (1) the rotor shall be turned to determine the value of the fundamental voltage. Without further turning of the rotor, the switch shall be changed to position (2) andthe null voltage shall be read. The maximum values of the voltage and of the fundamental component of the null voltage shall not exceed the values specified in 3.7.5, 3.7.6 and 3.7.7. The value of the fundamental voltage shall have been corrected for the attenuation of the filter used.

4.10 Electrical Stability Q Test of 115 Volt Control Synchros.---With the unit set up as indicated under the electrical error measurement and during the course of the electrical error test, when the unit is at electrical zero, electrical 60, electrical 120, electrical 180, electrical 240, and electrical 300, the input voltage shall be varied by ± 5% and the shift of null shall be recorded. The electrical error shall not change by more than ± 0.25 minutes.

4.11 Sensitivity Q Test for 115 Volt Pigtail Synchros.---This test is to be performed only on units having pigtails instead of slip rings. The torque differential receivers shall be tested against two torque transmitters of the same size. The units shall be connected as follows: the rotors of both TX's shall be connected in parallel R1 to R1, R2 to R2; the stator of one TX shall be connected to the rotor of the TDR6S1, S1 to R1, S2 to R2, S3 to R3; the stator of the other TX shall be connected to the stator of the TDR6S1, S1 to S1, S2 to S2, S3 to S3. With the shaft of one TX clamped, rotate the shaft of the other TX thru an angle of 10'. The shaft of the differential under test shall move detectably as determined by means of a light beam one meter long.

4.12 Torque Gradient Qualification Test for 115 Volt Synchros.---

4.12.1 General.---The unit shall be energized under standard conditions and shall be at normal operating temperature. This may be effected by energizing the unit with normal excitation current for 2 hours prior to the test. Torques shall be applied to the unit by suspending weights from a thread attached to the rim of a torque pulley on the rotor shaft. Predetermined weights of equal increments shall be applied that will cause deflections of approximately 2, 4, 6, 8, 9 and 10 degrees in both a clock-wise and counter-clockwise direction and the corresponding deflections recorded. A torque-deflection curve shall be plotted as the straight line best fitted

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:51Z
Check the source to verify that this is the current version before use.

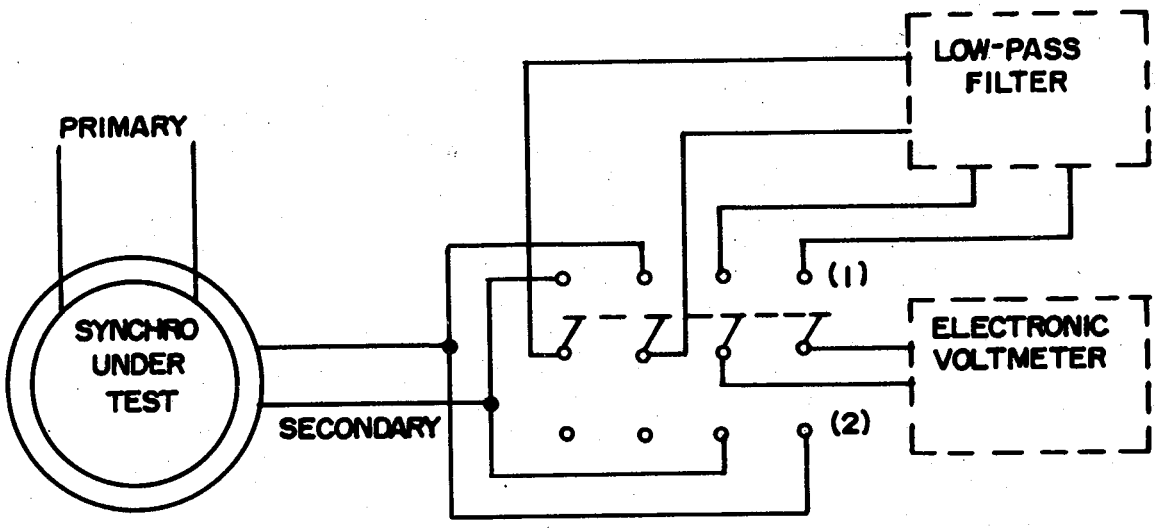


Table of Connections

Primary Supply			Required Null Voltage	
Test Unit	Across	Volts	Across Terminals	At Electrical Angles
CX TX	R1 R2	115.0	S1 S3 S3 S2 S2 S1	0°-180° 60°-240° 120°-300°
CDX	S2 (S1 S3)	78.0	R1 R3 R2 R1 R3 R2	0°-180° 60°-240° 120°-300°
CDX	S3 (S1 S2)	78.0	R2 R1 R3 R2 R1 R3	0°-180° 60°-240° 120°-300°
CDX	S1 (S2 S3)	78.0	R3 R2 R1 R3 R2 R1	0°-180° 60°-240° 120°-300°
CT	S2 (S1 S3) S1 (S2 S3) S3 (S1 S2)	78.0 78.0 78.0	R1 R2 R1 R2 R1 R2	0°-180° 60°-240° 120°-300°

Figure 11

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:51Z
Check the source to verify that this is the current version before use.

to the observed points and the torque gradient shall equal ½ the slope of this line and be expressed in terms of ounce-inches per degree.

4.12.2 Torque Gradient (Q) Test for Torque Transmitters and Receivers.---Torque transmitters and receivers shall be energized by supplying 115 volts at 60-cycles to the R1-R2 leads of the unit. Terminals S1 and S3 shall be connected together and the unit tested in accordance with 4.10.1. The torque gradient of torque transmitters shall conform to the limits given in 3.7.1. The torque gradient of torque receivers shall conform to the limits given in 3.7.2.

4.12.3 Torque Gradient (Q) Test for Torque Differential Transmitters and Receivers.---Torque differential transmitters and receivers shall be energized by supplying 78 volts at 60 cycles between terminals S2 and S1 which is connected to S3. Terminals R1, R3 shall be connected together and the unit tested in accordance with 4.10.1. The torque gradient of torque differential transmitters shall conform to the limits given in 3.7.3. The torque gradient of torque differential receivers shall conform to the limits given in 3.7.4.

4.13 Maximum Friction Torque (Q) Test for Driven Synchros (Transmitters and Control Transformers).---The purpose of this test is to determine whether the torque needed to turn the rotor of the synchro exceeds the value given in 3.7. This test shall be performed using an unbalanced dial or a pointer which, when mounted on the synchro shaft, shall exert a torque not greater than the maximum friction torque allowed in this specification. With the dial or pointer mounted on the shaft, the synchro stator shall be rotated at a rate not faster than 6 rpm while the shaft is in a horizontal position. At least two revolutions of the stator shall be made. The synchro shall be considered to have failed this test if the dial or pointer turns one revolution.

4.14 Radiation (Q) Test for all Synchros except 31TDR6S1.--- The rotor shall be rotated at 1150 ± 50 rpm and energized as indicated below. A Noise Meter shall be placed so that the brushes of the synchro are a distance of 3 ± .3 feet from the rod antenna of the noise meter. The noise level shall be measured at no less than two frequencies in each of the bands of the Noise Meter between 0.15 and 150 megacycles. The noise level shall be no more than 2.5 microvolts below 65 megacycles and no more than 5.0 microvolts above.

Type	Terminals	Voltage
Transmitters & Receivers	R1-R2	115 ± 5%
Differentials	R2, R1-R3	78 ± 5%
Transformers	R1-R2	57 ± 5%

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:51Z
Check the source to verify that this is the current version before use.

4.15 Terperature Rise Qualification Test.---

4.15.1 General.---The unit shall be tested when lying on its side on a heat insulating material shall as wood or ceramic. Prior to the temperature tests, the unit shall be maintained at 22°C ± 3° for not less than 24 hours. This ambient temperature shall be maintained throughout the test. The temperature rise of both the rotor and the stator of a unit shall be measured by the resistance change method. Measurements of resistance shall be recorded when equilibrium is attained. The following formula shall be used for calculating the temperature:

$$\text{Temperature Rise} \quad {}^{\circ}C = 254.5 \frac{(R_t - R22)}{R22}$$

where R_t = resistance of winding at the equilibrium temperature.
$R22$ = resistance of winding at 22°C

Schematic diagrams of the circuits to be used for transmitters and differentials are shown in figures 12 and 12a respectively.

4.15.2 Circuit.---Switch SW1 connects the bridge to the rotor or stator as desired. Switch SW2 is coupled mechanically to the bridge switch (not shown in figure) and closes when the bridge switch is closed. When switch SW2 is closed, the relay is energized disconnecting the synchro excitation voltage at the same time connecting the bridge to the rotor or the stator, depending upon the position of SW1. The relay has mercury contacts with a maximum variation in contact resistance of approximately plus or minus 0.004 ohm. This arrangement allows the bridge to be balanced in the conventional manner. Since the bridge switch and SW2 are closed just long enough to determine the deflection of the galvanometer needle, no significant cooling of the synchro occurs while the measurement is being made.

4.15.3 The synchro shall be kept at the electrical zero position throughout the test. The temperature rise test shall be done with no load on the synchro secondary and the temperature rise determined in accordance with 4.12.1. The temperature rise of any winding of the synchro shall not exceed the limits given in 3.7.

4.16 Spinning Qualification Test for 115 Volt Synchros.---The purpose of this test is to determine whether the rotor of a torque receiver or differential receiver will come to rest at the position of synchronism from any initial angular position without spinning. A spinning test dial as shown in Figure 15 shall be mounted on the rotor. The torque transmitter (or transmitters) shall be locked on electrical zero position. The rotor of the unit under test shall be displaced ± 175 ± 4° from electrical zero and the units energized with 130 volts at 60 cycles. This procedure shall be performed 5 times for a clockwise displacement and 5 times for a counter-clockwise displacement. The synchro shall be considered to have failed the spinning test if it fails to synchronize in any of the ten trials.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:51Z
Check the source to verify that this is the current version before use.

4.17 Synchronizing Time Qualification Test for 115 Volt Synchros.---

4.17.1 General.---The unit under test shall be connected to the corresponding terminals of a torque transmitter (or transmitters) of the same size, the rotor (or rotors) of which are locked. A test dial as shown in Figure 15 shall be mounted on the rotor of the unit to be tested. The synchronizing time is defined as the length of time required for the receiver rotor to synchronize and remain in synchronism within 0.1° after energizing the system at the specified voltage and frequency. The synchronizing time shall be measured for initial displacements of ± 30° and ± 175° by means of an oscillograph. The synchronizing time shall be recorded as the average of three successive tests. The energizing voltage shall remain 115 ± 3 volts throughout the test.

4.17.2 Synchronizing Time (Q) Test for Torque Receivers.---Receivers shall be energized by supplying 115 volts at 60 cycles to the R1-R2 leads of both units and tested in accordance with 4.14.1. The synchronizing time of torque receivers shall not exceed 1 second for a 30° displacement and 2 seconds for 175° displacement.

4.17.3 Synchronizing Time (Q) Test for Torque Differential Receivers.---Torque differential receivers shall be energized by supplying 115 volts at 60 cycles between terminals R1-R2 of both transmitters and tested in accordance with 4.17.1 The synchronizing time of torque differential receivers shall not exceed 1 second for a 30° displacement and 2 seconds for a 175° displacement.

4.18 High Potential and Insulation Resistance Qualification Test.---Synchro units shall pass a high voltage test of 900 ± 5% - 30 volts (rms) applied from each winding to the frame and from winding to winding. In all high voltage tests, the voltages shall be raised slowly to the specified value and held at that value for one minute. This test may be repeated as often as required. The peak voltage of the supply throughout this test shall not exceed 1.5 times the specified rms voltage. Immediately after satisfactorily understanding the high potential test the insulation resistance of the electrical circuits shall be measured by means of a megohm bridge at 400 volts D.C. between each winding and the case and between the primary and secondary windings. These tests shall be performed at -55 ± 5°C, 22 ± 5°C and 125 ± 5°C for all synchros. The insulation resistance shall be greater than 10 megohms at -55 ± 5°C and 22 ± 5°C. At 125 ± 5°C, the insulation resistance shall be greater than 4 megohms. The high voltage for this test may be either 60 or 400 cycles. The relative humidity for these tests shall not be less than 50%.

4.19 Dielectric Qualification Test.---Each synchro shall have twice the rated voltage ± 1% at twice the rated frequency ± 1% impressed upon its primary winding, with secondaries open. The voltage shall be raised slowly to the specified value and held at that value for one minute. The rotor shall be rotated at less than 2 rpm making at least

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:51Z
Check the source to verify that this is the current version before use.

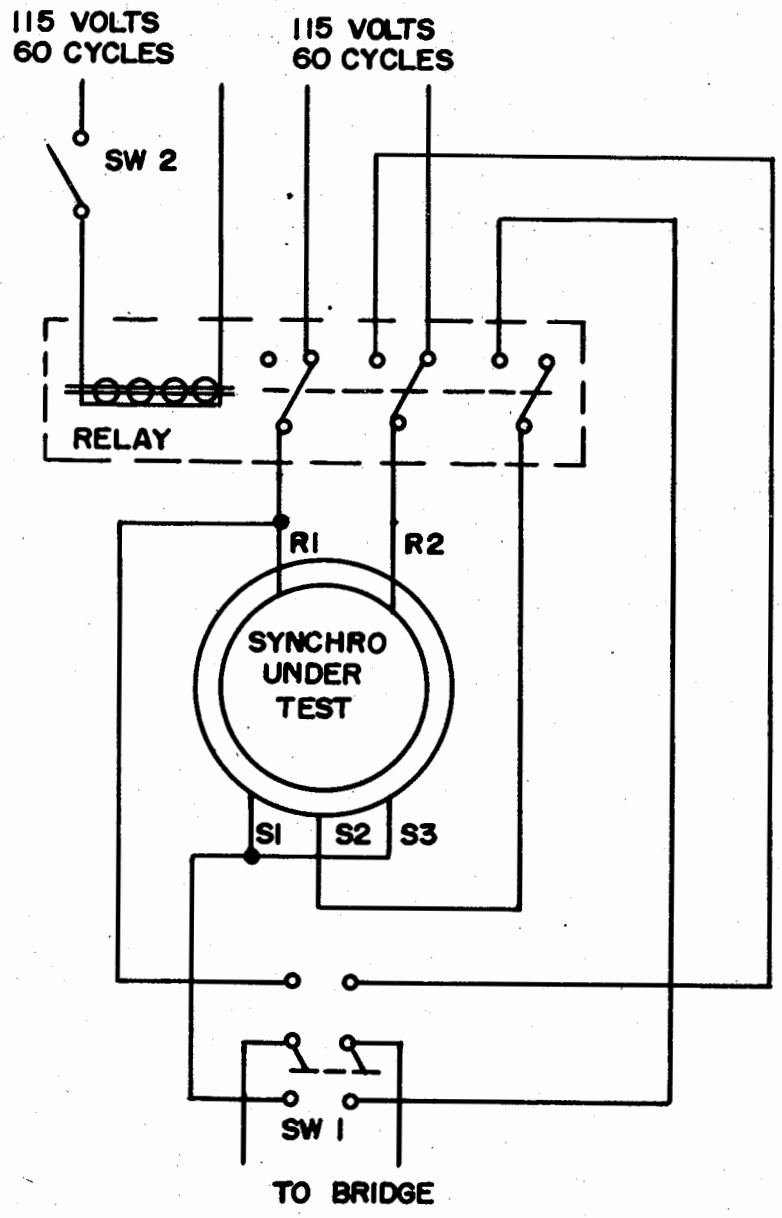


Figure 12

Circuit for measuring resistance changes of torque transmitters, torque receivers, and control transmitters.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:51Z
Check the source to verify that this is the current version before use.

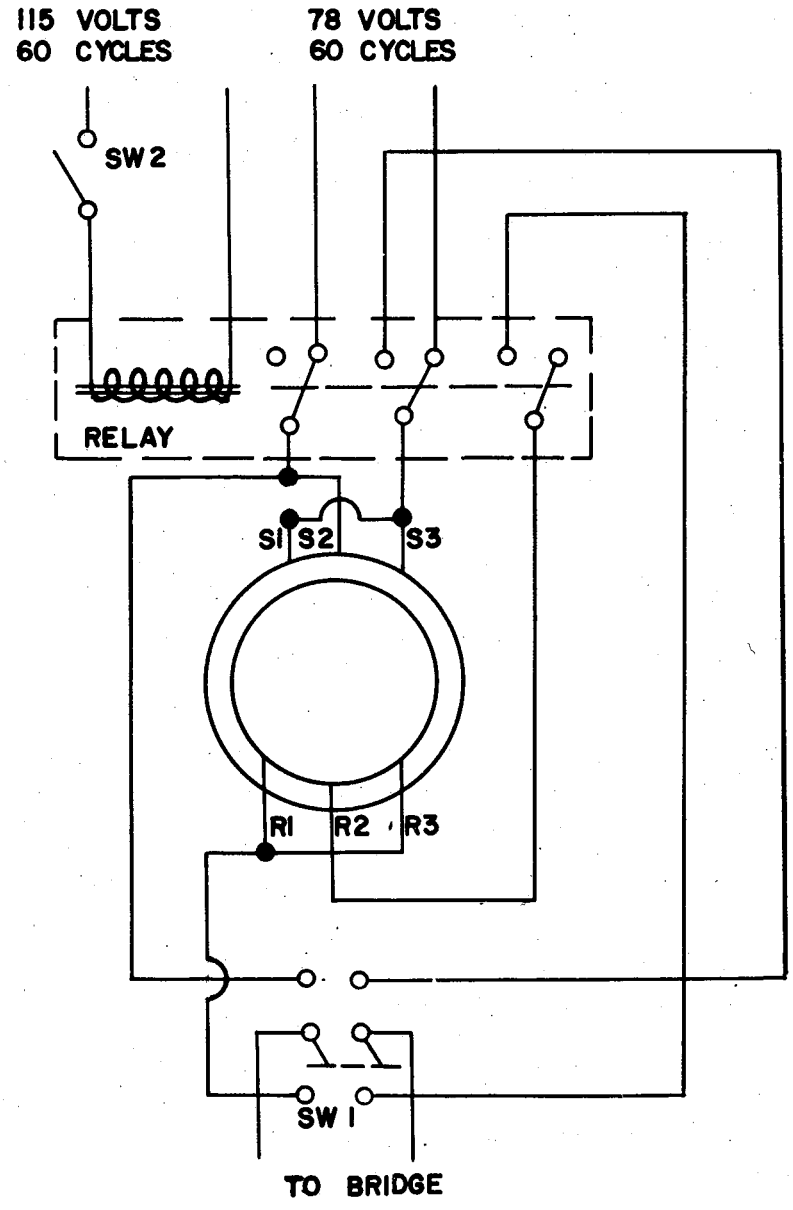


Figure 12a

Circuit for measuring resistance changes of torque differential transmitters, torque differential receivers and control differential transmitters.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:51Z
Check the source to verify that this is the current version before use.

one revolution throughout the period of the test. There shall be no indication of insulation breakdown as exhibited by a tendency of the rotor to seek a preferred position. Such a tendency may be accomplished by sporadic changes in the primary current. These tests shall be performed at -55 ± 5°C, and 125 ± 5°C for all synchros except the 31TDR6S1. The leads on the 31TDR6S1 shall be disconnected and test performed in accordance with above.

4.20 Vibration Qualification Test.--Either of the two methods described below may be used.

4.20.1 The unit to be tested shall be securely clamped to the table of a vibrator with its axis horizontal, and vibrated for a total of 24 hours in three mutually perpendicular directions, one being parallel to the axis of the rotor. During vibrations the table and synchro shall move in a plane defined by two of these principal directions; and the vibration in each plane shall be continued for 8 hours. In each plane, the frequency of vibration shall be held constant for one hour at each of the frequencies 800, 900, 1000, 1200, 1300, 1400 and 1500 cycles per minute. At all frequencies, the amplitude of vibration shall be 0.030 inch (0.060 inch total displacement).

4.20.2 The synchro shall be mounted rigidly to a vibration stand with the shaft vertical, with its extended end at the bottom and free to rotate. The synchro shall be vibrated simultaneously in the vertical and horizontal direction with a sinusoidal amplitude for each component of 0.10 ± 0.02 inches. The frequency of vibration shall vary linearly from 10 cps to 60 cps with a period of one minute. The total time of vibration shall be 8 hours ± 15 minutes.

4.20.3 The synchro shall not be damaged by this test, nor shall any of the parts loosen as a result of the test. After the test the rotor and bearings shall be removed and examined. The presence of flakes of coating, loose metal chips or other particles of foreign matter shall be cause for rejection. The synchro shall be capable of passing the qualification accuracy test after vibration.

4.21 Shock Qualification Test.---Each unit shall be mounted in a special shock test fixture shown in Bureau of Ordnance Sketch No. 95216 and shall be subjected to six (6) blows of 2000 ft. lb. with the high intensity shock machine. Three (3) blows shall be struck in the vertical and three (3) blows in the horizontal direction. The synchro shall be energized in the usual manner during the test. The special shock fixture shall be bolted to a standard 34" x ½" thick mounting plate which is fastened to the shock testing machine anvil plate by two (2) 4" channels. The test fixture shall be mounted so that a line between the brushes makes an angle of approximately 45° with the mounting plate. All receivers shall have a standard type dial (see Bureau of Ordnance Sketch No. 81883) or equivalent load mounted on the shaft during the test. The unit shall not be damaged or deranged by this test. The errors in static accuracy at the end of the test shall not exceed the values given in 3.7. The unit shall also pass the high potential, insulation resistance, and dielectric tests performed at room temperature. For a description of the shock testing machine see MIL-S-901.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:51Z
Check the source to verify that this is the current version before use.

4.22 Endurance Qualification Test.---

4.22.1 Driven Units.---

4.22.1.1 General.---All synchros except 31TDR6S1 shall be given a continuous endurance run of 1000 ± 10 hours at 1150 ± 50 rpm while energized at rated voltage and frequency. Type 31TDR6S1 shall be given a continuous endurance run of 1000 ± 10 hours at 96 ± 4 sinusoidal oscillations per minute. Amplitude of oscillation shall be 15° (peak to peak - 30°) from the electrical zero position. The first 64 hours of this test shall be made in an ambient temperature of minus 25°C with the shaft of the unit horizontal. The next 96 hours of the test shall be made in an ambient of 65°C and the unit shall be operated in each of the following positions for 24 hours:

(1) Shaft vertical, upward.
(2) Shaft inclined 45° upward.
(3) Shaft inclined 45° downward.
(4) Shaft vertical, downward.

The remaining time of the test shall be consumed by running the unit with the shaft horizontal and at room temperature. Static accuracy qualification tests shall be made both before and after the endurance test. The units shall then be disassembled and checked for evidence of undue wear, deterioration of windings and deposits of distillates.

4.22.1.2 Endurance (Q) Test of Transmitters and Torque Receivers.---Transmitters shall be energized by supplying 115 volts ± 1% at 60 cycles to the R1-R2 terminals. The rotor shall be turned mechanically at the speed required in 4.19.1.1.

4.22.1.3 Endurance (Q) Test of Control Transformers.---Control Transformers shall be energized by applying 78 volts ± 1% at 60 cycles between terminals S2 and S1 which is connected to S3. The rotor shall be driven mechanically as required in 4.19.1.1. A load of 13,000 ohms shall be connected across R1-R2.

4.22.1.4 Endurance (Q) Test of Differential Transmitters and Torque Differential Receivers.---Differential transmitters shall be energized by spplying 78 volts ± 1% at 60 cycle between terminals S2 and S1 which is connected to S3. The rotor shall be turned mechanically at the speed required in 4.19.1.1.

4.22.1.5 Endurance (Q) Test of 31TDR6S1.---The 31TDR6S1 shall be energized by applying 78 volts ± 1% at 60 cycles between terminals S2 and S1 which is connected to S3. The rotor shall be oscillated mechanically at the speed required in 4.22.1.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:51Z
Check the source to verify that this is the current version before use.

4.23 Sequence of Vibration, Shock, and Endurance Tests.---The laboratory accomplishing the type approval tests may perform the specified retests after each test described in 4.17 (Vibration), 4.18 (Shock) and 4.19 (Endurance) or after any combination of these tests.

4.24 Humidity Qualification Test.---

4.24.1 Humidity Test.---The synchro shall be subjected to the following humidity cycling test (specified in MIL-E-5272) Procedure I):

6 hours at 160°F (71°C) and 95% to 100% relative humidity.
16 hours during which the temperature drops to 100°F (38°C) or less

repeat cycle for 360 hours (15 cycles). After the test the rotor and the bearings shall be removed and examined for evidence of rust or corrosion. The following shall be considered causes for rejection:

a. Any visible rust on ferrous parts except the external surfaces of bearing races.
b. Any blistering of the coating on the ferrous parts which indicates the presence of rusting under the coat.
c. Any corrosion which would result in a failure of the synchro.
d. Any varnish condensates on the bearings. The bearing shields shall be removed for this examination. Rust on the external surfaces of the bearing races shall not be considered cause for rejection.

4.25 Mechanical Inspection (Q).---The synchros shall be carefully checked to determine conformity of the dimensions and furnishes with the applicable installation drawings listed in 2.1 of this specification. The synchros shall also be examined for workmanship, cleanliness, lubrication and the tightness of all fastenings. All terminal block screws shall be tight prior to each type approval test.

4.26 Electrical Error Inspection Test for 115 Volt Synchros.---The inspector may use either the (Q) test described in 4.5 or the tests described below. If the tests below are used, one synchro in each group of ten (10) is to be tested also in accordance with the qualification test of 4.5. Failure of the two results to agree within $\pm$ 2 minutes at all points shall require all units of the group to be tested for qualification (paragraph 4.5). The results of the latter test shall apply. If ten consecutive groups indicate agreement between the two tests to the accuracy indicated, the group sample may be enlarged to twenty-five (25). Failure to agree on any test shall result in reversion to ten (10) units per group.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:51Z
Check the source to verify that this is the current version before use.

4.26.1 Electrical Error (I) Test of 115 Volt Transmitters and Torque Receivers.---Transmitters and torque receivers shall be tested against control transformers calibrated by the proportional voltage method as described in 4.5.4. The synchro under test shall be energized by applying 115 volts 60 cycles to the rotor. Its stator leads shall be connected to the stator leads of the calibrated control transformer in the following manner: S1 to S3, S3 to S1 and S2 to S2, and its rotor shall be coupled mechanically to the rotor of the control transformer so that both units are at electrical zero together. The design of the entire coupling fixture must be approved by the contracting officer. The rotor voltage of the control transformer is a measure of the static error of the synchro. This voltage shall be recorded for a complete revolution by means of a recording voltmeter. The electrical error record shall be corrected if necessary to account for the errors of the calibrated control transformer. The errors of 115 volt control transmitters shall not exceed those given in 3.7.5. The errors of 115 volt torque transmitters shall not exceed those given in 3.7.1. The errors of 115 volt torque receivers shall not exceed those given in 3.7.2. The error is defined as the rotor position minus the corrected position of the calibrated unit. A positive maximum of the 2nd harmonic of the error shall occur in the 0° to 120° sector provided there is a 2nd harmonic error of more than 3 minutes maximum value. The position of electrical zero shall be indicated on the record.

4.26.2 Electrical Error (I) Test of 115 Volt Differential Transmitters and Torque Differential Receivers.---Except for the 31TDR6S1, the static accuracy of both the rotor and the stator shall be determined.

4.26.2.1 Rotors.---Rotors of 115 volt differential transmitters and torque differential receivers shall be tested against control transformers calibrated by the proportional voltage method as described in 4.5.4. The unit under test shall be energized by applying 78 volts 60 cycles between terminals S2 and S1 which is connected to S3. Its rotor leads shall be connected to the stator leads of the calibrated control transformer in the following manner: R1 to S1, R3 to S3, and R2 to S2. Its rotor shall be coupled mechanically to the rotor of the control transformer so that both units are at electrical zero together. The electrical errors, corrected if necessary for the calibration of the control transformer, shall be determined by means of a recording voltmeter as described in 4.23.1. The errors for control differential transmitters shall not exceed those listed in 3.7.3. The errors for torque differential transmitters shall not exceed those listed in 3.7.6. The errors of torque differential receivers shall not exceed those listed in 3.7.7. The position of electrical zero shall be indicated on the record. The error is defined as the rotor position minus the corrected position of the calibrated unit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:51Z
Check the source to verify that this is the current version before use.

4.26.2.2 Stators.---Stators of 115 volt differential transmitters and torque differential receivers shall be tested against transmitters calibrated by the proportional voltage method as described in 4.5.4. The transmitter shall be energized by applying 115 volts 60 cycles to its rotor. The stator leads of the differential transmitter or receiver under test shall be connected to the stator of the calibrated transmitter in the following manner: S1 to S3, S3 to S1, and S2 to S2, and its rotor shall be coupled mechanically to the transmitter rotor so that both units are at electrical zero together. The electrical errors, corrected if necessary for the calibration of the transmitter, shall be determined by means of a recording voltmeter as described in 4.26.1. The errors for control units shall not exceed those listed in 3.7.3. The errors for torque differential transmitters shall not exceed those listed in 3.7.6. The errors of torque differential receivers shall not exceed those listed in 3.7.7. The error is defined as the rotor position minus the corrected position of the calibrated unit. A positive maximum of the 2nd harmonic of the error shall occur in the 0 to 120 degree sector provided there is a 2nd harmonic of error of more than 3 minutes maximum value. The position of electrical zero shall be indicated on the record.

4.26.3 Electric Error (I) Test of 115 Volt Control Transformers.--- Control transformers shall be tested against transmitters calibrated by the proportional voltage method as described in 4.5.3. The transmitter shall be energized by applying 115 volts 60 cycles to its rotor. The stator leads of the control transformer under test shall be connected to the stator leads of the calibrated transmitter in the following manner: S1 to S3, S3 to S1, and S2 to S2, and its rotor shall be coupled mechanically to the transmitter rotor so that both units are at electrical zero together. The rotor voltage of the control transformer is a measure of its error. The electric errors, corrected if necessary for the calibration of the transmitter, shall be determined by means of a recording voltmeter as described in 4.23.1. The errors of 115 volt control transformers shall not exceed those listed in 3.7.7. The error is defined as the rotor position minus the corrected position of the calibrated unit. A positive maximum of the 2nd harmonic of the error shall occur in the 0° to 120° sector provided there is a 2nd harmonic of error of more than 3 minutes maximum value. The electrical zero position shall be indicated on the record.

4.27 Sensitivity (I) Test for 115 Volt Pigtail Synchros.---This test shall be performed only on units having pigtails instead of slip rings. The torque differential receivers shall be tested against two torque transmitters of the same size. The units shall be connected as follows: the rotor of both TX's shall be connected in parallel

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:51Z
Check the source to verify that this is the current version before use.

R1 to R1, R2 to R2; the stator of one TX shall be connected to the rotor of the TDR6S1, S1 to R1, S2 to R2, S3 to R3; the stator of the other TX shall be connected to the stator of the TDR6S1, S1 to S1, S2 to S2, S3 to S3. With the shaft of one TX clamped, rotate the shaft of the other TX thru an angle of 10'. The shaft of the differential under test shall move detectably as determined by means of a light beam one meter long.

4.28 Receiver Error Inspection Test for 115 Volt Synchros.---

4.28.1 Receiver Error (I) Test of 115 Volt Torque Receivers.---Torque receivers shall be tested by means of the Synchro Tester Mark 1 Mod 1 or an equivalent unit which may be supplied by the contracting agency. The static errors of receivers shall not exceed those listed in 3.7.2.

4.28.2 Receiver Error (I) Test of 115 Volt Torque Differential Receivers.---Torque Differential receivers shall be tested by means of the Synchro Tester Mark 1 Mod 1 or an equivalent unit which may be supplied by the contracting agency and a second calibrated transmitter which is locked on electrical zero. The errors of torque differential receivers shall not exceed those listed in 3.7.4.

4.29 Transformation Ratio Inspection Test for 115 Volt Synchros.--- By the transformation ratio of a synchro is meant the ratio of the no-load maximum secondary voltage to the supply voltage applied to the primary. For this test the voltage measuring device, shall not alter the open circuit secondary voltage by more than 0.1%.

4.29.1 115 Volt Transmitters and Receivers.---The transformation ratio of 115 volt transmitters and receivers (which should be 90/115= 0.783) shall be obtained by applying 115 volts $\pm$ 1% at 60 cycles $\pm$ 1% across synchro terminals R1-R2 and turning the rotor until voltage E(S13) is a maximum. The transformation ratio shall be 0.783 $\pm$ 1%.

4.29.2 115 Volt Differential Transmitters and Torque Differential Receivers.---The transformation ratio of 115 volts differential transmitters and torque differential receivers (which should be 90/78 = 1.154) shall be obtained by applying 78 volts $\pm$ 1% at 60 cycles $\pm$ 1% across terminals S2 and S1 which is connected to S3. The rotor shall be turned until voltage E(R13) is a maximum. This can only be done prior to final assembly for a 31TDR6S1. The transformation ratio shall be 1.154 $\pm$ 1%.

4.29.3 115 Volt Control Transformers.---The transformation ratio of 115 volt control transformers (which should be 57.3/78 = 0.735) shall be obtained by applying 78 volts $\pm$ 1% at 60 cycles $\pm$ 1% across terminals S2 and S1 which is connected to S3. The rotor shall be turned until voltage E(R12) is a maximum. The transformation ratio shall be 0.735 $\pm$ 1%.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:51Z
Check the source to verify that this is the current version before use.

4.30 Open Circuit Current and Power (I) of 115 Volt Synchros.--- The input current and power to the synchro shall be measured with the synchro energized at rated voltage and frequency and with the secondary leads open. For differential synchros and control transformers the applied voltage is 78 volts, at 60 cycles, which shall be applied across terminals S2 and S1 which is connected to S3. The measurements shall be made with the synchro at room temperature. The primary current and power shall not exceed the values given in 3.7. The open circuit current or power shall not vary by more than 2 percent as the rotor is turned through a complete revolution.

4.31 Null Voltage Inspection Test for 115 Volt Synchros.---The null voltage and the fundamental component of the null voltage shall be determined by employing the circuit of Figure 13. These voltages shall be read on a vacuum tube voltmeter indicating the average value of the voltage wave in terms of the rms value of an equivalent sine wave. The synchro shall be connected as indicated in the table of Figure 13. The null voltage readings shall be made with the synchro at the electrical angles shown in the table. With the switch in position (1) the rotor shall be turned to determine the value of the fundamental voltage. Without further turning of the rotor, the switch shall be changed to position (2) and the null voltage shall be read. The maximum values of the null voltage and of the fundamental component of the null voltage shall not exceed the values specified in, 3.7.5, 3.7.6, and 3.7.7. The value of the fundamental voltage shall have been corrected for the attenuation of the filter used.

4.32 Spinning Inspection Test for 115 Volt Synchros.---The purpose of this test is to determine whether the rotor of a torque receiver or differential receiver will come to rest at the position of synchronism from any initial angular position without spinning. A spinning test dial as shown in Figure 15 shall be mounted on the rotor. The torque transmitter (or transmitters) shall be locked on electrical zero position. The rotor of the unit under test shall be displaced $\pm$ 179° from electrical zero and the units energized with 130 volts at 60 cycles. This procedure shall be performed 5 times for a clockwise displacement and 5 times for a counter-clockwise displacement. The synchro shall be considered to have failed the spinning test if it fails to synchronize in any of the ten trials.

4.33 Synchronizing Time Inspection Test for 115 Volt Synchros.---

4.33.1 General.---The unit under test shall be connected to the corresponding terminals of a torque transmitter (or transmitters) of the same size. A test dial as shown in Figure 15 shall be mounted on the rotor of the unit. The synchronizing time is defined as the length of time required for the receiver rotor to synchronize and remain in synchronism within 0.1° after energizing the system at the specified voltage and frequency. The synchronizing time shall be measured for initial displacements of $\pm$ 30° and $\pm$ 175° by means of an oscillograph. The synchronizing time shall be recorded as the average of three successive tests. The energizing voltage shall remain 115 $\pm$ 3 volts throughout the test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:51Z
Check the source to verify that this is the current version before use.

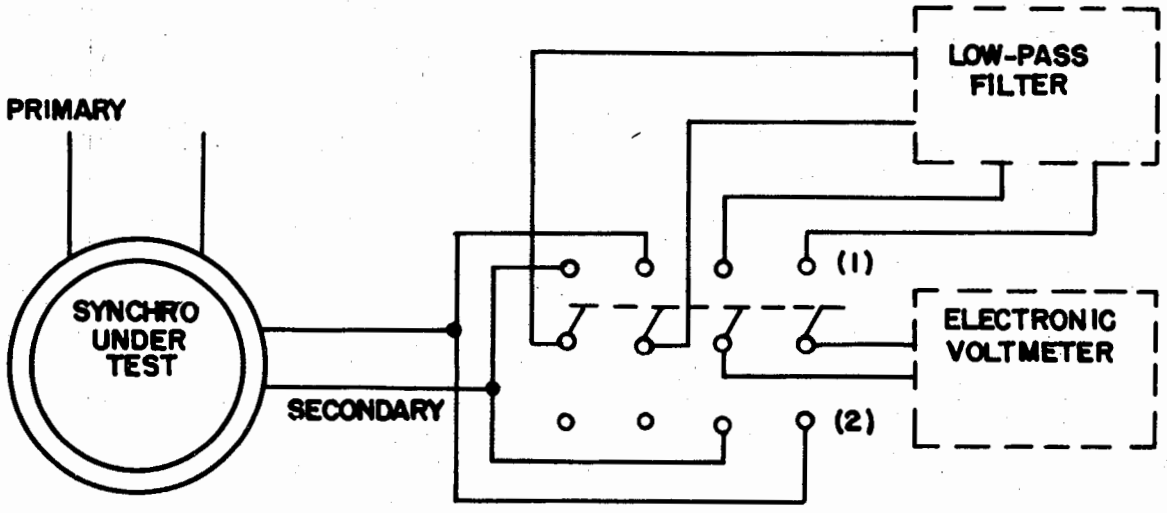


Table of Connections

	Primary Supply		Required Null Voltage	
Test Unit	Across	Volts	Across Terminals	At Rotor Angles (Approx.)
CX TX	R1 R2	115.0	S1 S3 S3 S2 S2 S1	0°-180° 60°-240° 120°-300°
CDX	S2 (S1 S3)	78.0	R1 R3 R2 R1 R3 R2	0°-180° 60°-240° 120°-300°
CDX	S3 (S1 S2)	78.0	R2 R1 R3 R2 R1 R3	0°-180° 60°-240° 120°-300°
CDX	S1 (S2 S3)	78.0	R3 R2 R1 R3 R2 R1	0°-180° 60°-240° 120°-300°
CT	S2 (S1 S3) S1 (S2 S3) S3 (S1 S2)	78.0 78.0 78.0	R1 R2 R1 R2 R1 R2	0°-180° 60°-240° 120°-300°

Figure 13

Null Voltage Test Circuit for 115 Volt Synchros

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:51Z
Check the source to verify that this is the current version before use.

4.32.2 Synchronizing Time (I) Test for 115 Volt Receivers.---Receivers shall be energized by supplying 115 volts at 60 cycles to the R1-R2 leads of both units and tested in accordance with 4.29.1. The synchronizing time of torque receivers shall not exceed 1 second for a 30° displacement and 2 seconds for a 175° displacement.

4.33.3 Synchronizing Time (I) Test for 115 Volt Torque Differential Receivers.---Torque differential receivers shall be energized by supplying 115 volts at 60 cycles between terminals R1-R2 of both transmitters and tested in accordance with 4.33.1. The synchronizing time of torque differential receivers shall not exceed 1 second for a 30° displacement and 2 seconds for a 175° displacement.

4.34 Maximum Friction Torque (I) Test for 115 Volt Driven Synchros (Transmitters and Control Transformers).---The purpose of this test is to determine whether the torque needed to turn the rotor of the synchro exceeds the value given in 3.7. This test shall be performed using an unbalanced dial or a pointer which, when mounted on the synchro shaft, shall exert a torque not greater than the maximum friction torque allowed in this specification. With the dial or pointer mounted on the shaft, the synchro stator shall be rotated at a rate not faster than 6 rpm while the shaft is in a horizontal position. At least two revolutions of the stator shall be made. The synchro shall be considered to have failed this test if the dial or pointer turns one revolution.

4.35 High Potential and Insulation Resistance Inspection Test for 115 Volt Synchros.---Synchro units shall pass a high voltage test of 900 $\pm$ 0 - 30 volts (rms) applied from each winding to the frame and from winding to winding. In all high voltage tests, the voltages shall be raised slowly to the specified value and held at that value for one minute. This test may be repeated as often as required. The peak voltage of the supply throughout this test shall not exceed 1.5 times the specified rms voltage. Immediately after satisfactorily withstanding the high potential test the insulation resistance of the electrical circuits shall be measured by means of a megohm bridge at 500 volts D.C. between each winding and the case and between the primary and secondary windings. These tests shall be performed at -55 $\pm$ 5°C, 22 $\pm$ 5°C and 125 $\pm$ 5°C for all synchros. The insulation resistance shall be greater than 10 megohms at -55 $\pm$ 5°C and 22 $\pm$ 5°C. At 125 $\pm$ 5°C, the insulation resistance shall be greater than 4 megohms.

4.36 Dielectric Inspection Test for 115 Volt Synchros.---Each synchro shall have twice the rated voltage at twice the rated frequency impressed upon its primary winding, with secondaries open. The voltage shall be raised slowly to the specified value and held at that value for one minute. The rotor shall be rotated at less than 2 rpm making at least one revolution throughout the period of the test. There shall be no indication of insulation breakdown as exhibited by a tendency of the rotor to seek a preferred position. Such a tendency may be accompanied by sporadic changes in the primary current. These tests shall be performed at -55 $\pm$ 5°C, and 125 $\pm$ 5°C for all synchros except the 31TRD6S1. The dielectric inspection test for type 31TDR6S1 shall be performed prior to final assembly.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:51Z
Check the source to verify that this is the current version before use.

4.37 Mechanical Inspection (I) for 115 Volt Synchros.---The synchros shall be carefully checked to determine conformity of the dimensions and finishes with the applicable installation drawings listed in 2.1 of this specification. The synchros shall also be examined for workmanship, cleanliness, lubrication and the tightness of all fastenings.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:51Z
Check the source to verify that this is the current version before use.

Adjust condenser "A" for zero meter reading with 60 cycle input.
Adjust condenser "B" for zero meter reading with 400 cycle input.

Component Tolerances 20% unless otherwise noted				
R1C1 = .00216 ± 2%,	.1M	R1	.5M	Reference voltage input
R2C2 = .000325 ± 2%,	.1M	R2	.5M	75 to 120 volts, 60 or 400 cps, in phase with primary line voltage.

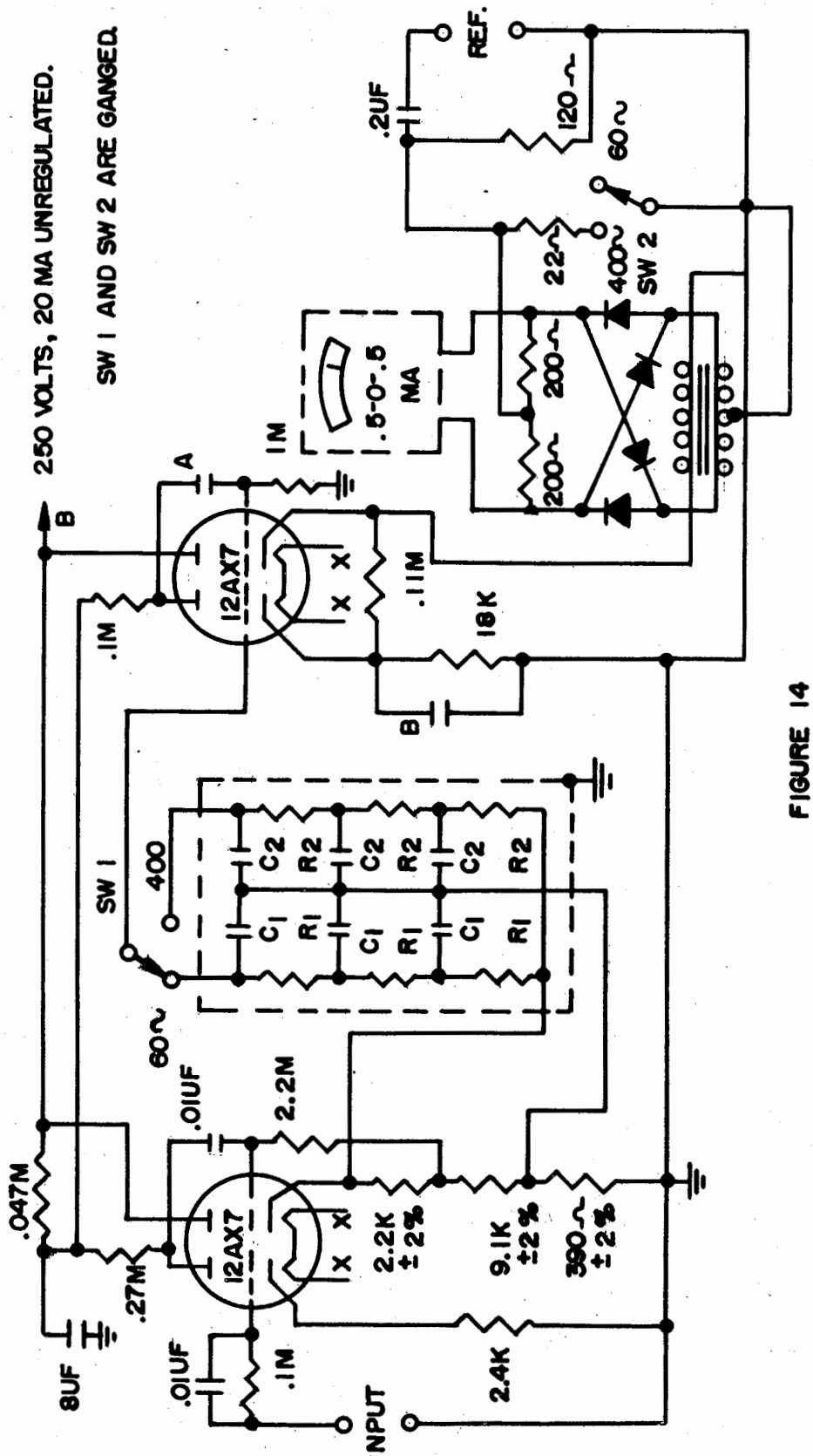
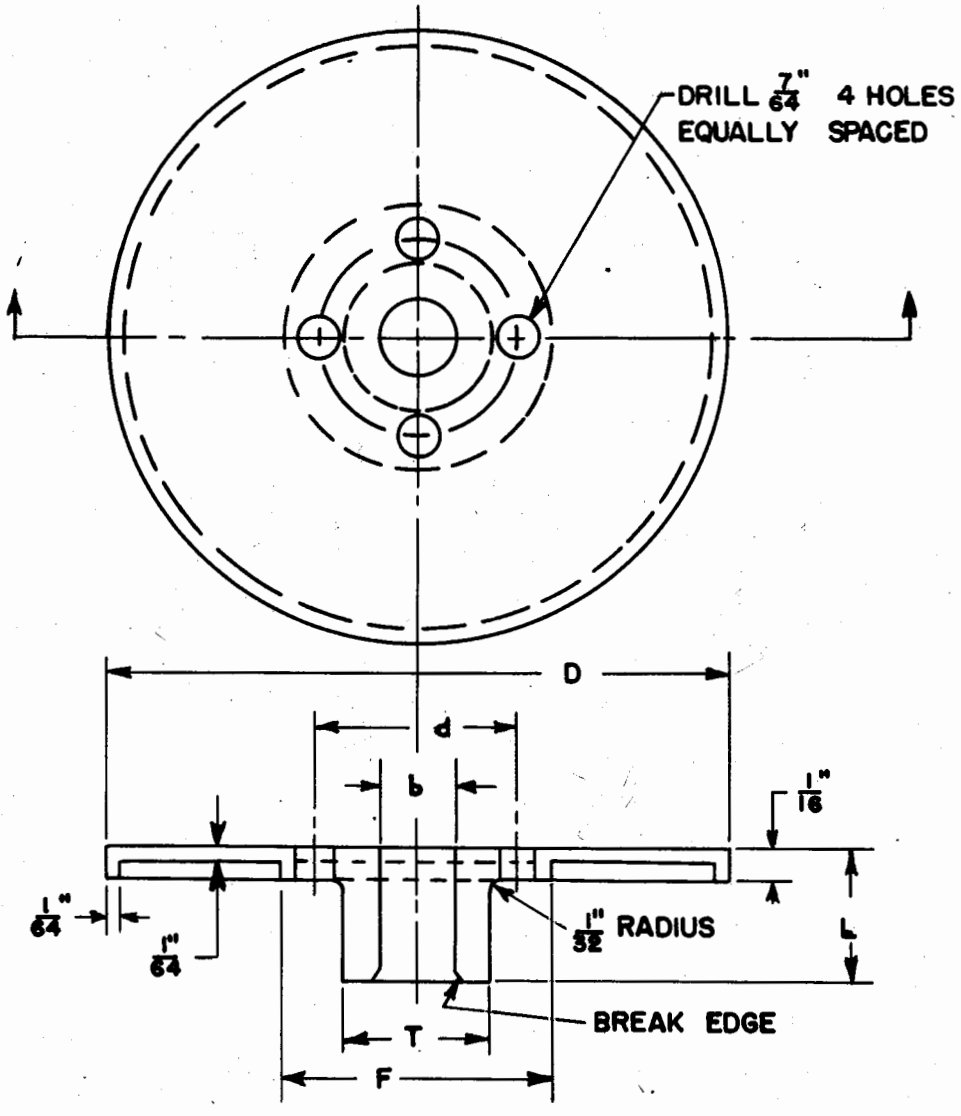


FIGURE 14

NULL METER

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:51Z
Check the source to verify that this is the current version before use.

Size	D	d	b	L	T	f
23	2.67	.437 ± .005	.2410 ± .0004 - .0000	.350 ± .005 - .000	.315 ± .005 - .000	.631 ± .005 - .000
31	3.50	.437 ± .005	.2410 ± .0004 - .0000	.350 ± .005 - .000	.315 ± .005 - .000	.631 ± .005 - .000



MATERIAL: ALUMINUM

SYNCHRO DIAL FOR SYNCHRONIZING TEST AND SPINNING TEST

FIGURE 15

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:51Z
Check the source to verify that this is the current version before use.

5. PREPARATION FOR DELIVERY

5.1 Packaging, Packing and Marking.- Each synchro shall be packaged, packed and marked in accordance with Specification MIL-P-12134.

6. NOTES

6.1 Invitations for bids and contracts should include the following provision covering guaranty:

"The contractor shall guarantee all synchros, including spare parts and accessory material, furnished by him, against any and all defects in materials and workmanship which are not detected prior to acceptance but which may develop within two years after delivery of the material, such defects to be corrected by and at the expense of the contractor."

NOTICE. - This specification, together with specifications and drawings pertaining to it and bearing a "Notice" of similar restrictions, is intended for use only in connection with procurement by the United States Government; and shall not be reproduced either wholly or in part except when authorized in connection with Government procurement, nor be used for any other purpose except when specifically authorized by the Chief of Ordnance.

CUSTODIAN:
Army - Ordnance Corps

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:51Z
Check the source to verify that this is the current version before use.